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“ΕΜΒΡΥΟΜΗΤΡΙΚΗ ΙΑΤΡΙΚΗ”

ΜΕΤΑΠΤΥΧΙΑΚΗ ΔΙΠΛΩΜΑΤΙΚΗ ΔΙΑΤΡΙΒΗ

ΘΕΜΑ

Συγκριτική αναδρομική μελέτη κεφαλουραίου μήκους
εμβρύων εξωσωματικής γονιμοποίησης μετά από μεταφορά
κατεψυγμένων έναντι φρέσκων βλαστοκύστεων

Υπό

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ΘΕΣΣΑΛΟΝΙΚΗ 2025

Δηλώνω υπεύθυνα ότι η διπλωματική εργασία έχει συνταχθεί από εμένα και κανένα μέρος της δεν είναι αντιγραμμένο από έντυπες ή ηλεκτρονικές πηγές, μετάφραση από ξενόγλωσσες πηγές και αναπαραγωγή από εργασίες άλλων ερευνητών ή φοιτητών.

Όπου έχω βασιστεί σε ιδέες ή κείμενα άλλων, έχω προσπαθήσει με όλες μου τις δυνάμεις να το προσδιορίσω σαφώς μέσα από την καλή χρήση αναφορών ακολουθώντας την ακαδημαϊκή δεοντολογία.

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25/01/2025

ΕΥΧΑΡΙΣΤΙΕΣ

Θερμές ευχαριστίες στα μέλη της Τριμελούς Εξεταστικής Επιτροπής για την ανάθεση αυτού του εξαιρετικά ενδιαφέροντος θέματος.

Τον κ. Απόστολο Αθανασιάδη, Τακτικό Καθηγητή Μαιευτικής Γυναικολογίας και Εμβρυομητρικής Ιατρικής Α.Π.Θ., ευχαριστώ θερμά για την επιλογή του θέματος και την προτροπή του να παρακολουθήσω τα τακτικά ιατρεία προγεννητικής παρακολούθησης φυσιολογικών και κήσεων υψηλού κινδύνου.

Τον κ. Ιωάννη Τσακίριδη, Επίκουρο Καθηγητή Μαιευτικής Γυναικολογίας και Εμβρυομητρικής Ιατρικής Α.Π.Θ., ευχαριστώ θερμότατα για τη στατιστική ανάλυση των δεδομένων και επεξεργασία των αποτελεσμάτων.

Τον κ. Θεμιστοκλή Δαγκλή, Αναπληρωτή Καθηγητή Μαιευτικής Γυναικολογίας και Εμβρυομητρικής Ιατρικής Α.Π.Θ. ευχαριστώ θερμά για τα κατατοπιστικότατα θεωρητικά και πρακτικά μαθήματα και τις επεξηγήσεις επί του θέματος στα πλαίσια του εκπαιδευτικού μεταπτυχιακού προγράμματος.

ΠΡΟΛΟΓΟΣ

Το αντικείμενο της παρούσας διπλωματικής εργασίας αφορά στην αναδρομική συγκριτική εκτίμηση του κεφαλουραίου μήκους εμβρύων εξωσωματικής γονιμοποίησης των οποίων η ενδομήτρια μεταφορά πραγματοποιήθηκε είτε άμεσα, είτε σε δεύτερο χρόνο μετά από διαδικασία ψύξης-απόψυξης.

Όπως έχει ήδη παρατηρηθεί σε διεθνείς μελέτες και επιβεβαιώθηκε σ' αυτή τη μελέτη, τα κατεψυγμένα έμβρυα παρουσιάζουν μεγαλύτερο κεφαλουραίο μήκος μεταξύ 11ης – 14ης εβδομάδας κύησης σε σχέση με εκείνα που μεταφέρθηκαν ως φρέσκα. Το εύρημα αυτό θεωρείται αποτέλεσμα πολυπαραγοντικής διαδικασίας που σχετίζεται με πολλαπλές παραμέτρους των πρωτοκόλλων εξωσωματικής γονιμοποίησης με ή χωρίς κατάψυξη εμβρύων, καθώς και με εγγενή χαρακτηριστικά των ασθενών όπως η ανατομία της μήτρας και η μητρική καρδιοαγγειακή λειτουργία πρό κύησης. Ο συνδυασμός αυτών των παραγόντων επιδρά στο ορμονικό προφίλ και στην αιμάτωση της μήτρας επηρεάζοντας την ανάπτυξη της εμβρυοπλακουντιακής μονάδας, ενώ περιγράφονται και επιγενετικές αλλαγές που συνεπικουρούν στα ευρήματα, και που πιθανώς παρουσιάζουν επιπλέον αντίκτυπο μακροπρόθεσμα στην υγεία του παιδιού. Όλοι αυτοί οι παράγοντες και η αλληλεπίδρασή τους χρήζουν περαιτέρω διερεύνησης.

Η βιομετρία του κεφαλουραίου μήκους κατά το πρώτο τρίμηνο κύησης συνάδει με μεγαλύτερα βάρη γέννησης σε περιπτώσεις κατεψυγμένων σε σύγκριση με φρέσκων εμβρύων. Επομένως, η εκτίμηση των διαφορών του κεφαλουραίου μήκους μεταξύ των δύο αυτών ομάδων μπορεί να συμβάλλει στην κατανόηση των παθοφυσιολογικών μηχανισμών που διαφοροποιούν την αναπτυξιακή πορεία τους.

Στο πρώτο γενικό μέρος της μελέτης αναφέρεται η γεωμετρική αύξηση στη συχνότητα εφαρμογής εξωσωματικής γονιμοποίησης κατά την τελευταία εικοσαετία και γίνεται μια αδρή περιγραφή των βασικών μεθόδων καθώς και των αντίστοιχων διαφορών που παρατηρούνται στην ενδομήτρια ανάπτυξη του εμβρύου και στο βάρος γέννησης. Γίνεται επιγραμματική αναφορά στις πιθανές μητρικές και εμβρυικές επιπλοκές και στους γενεσιουργούς τους μηχανισμούς.

Στο δεύτερο ειδικό μέρος παρατίθενται τα καταγραφικά δεδομένα 3,082 μονήρων και διδύμων κύησης εξωσωματικής γονιμοποίησης που αντιστοιχούσαν σε 4,044 έμβρυα. Περιγράφεται η μεθοδολογία στατιστικής ανάλυσης και τα αποτελέσματα. Γίνεται σύγκριση κεφαλουραίου μήκους μεταξύ κατεψυγμένων και φρέσκων εμβρύων καθώς και μεταξύ δύο

υποομάδων : μεταξύ εμβρύων από ωάρια δότριας και από αυτόλογα ωάρια, και μεταξύ διδύμων και μονήρων. Στη συζήτηση προσεγγίζεται η ερμηνεία των αποτελεσμάτων με βάση τα δεδομένα της διεθνούς βιβλιογραφίας. Με βάση αυτά, συμπεραίνεται η αξιολόγηση του κεφαλουραίου μήκους ως έμμεσου προγνωστικού δείκτη της εμβρυικής ανάπτυξης.

Στο τελευταίο τμήμα της εργασίας παρατίθεται η περίληψη και οι λέξεις-κλειδιά στην Ελληνική και Αγγλική γλώσσα καθώς και η βιβλιογραφία.

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ABBREVIATIONS

ART	Assisted Reproductive Technology
IVF	In Vitro Fertilization
ICSI	Intracytoplasmic Sperm Injection
OD	Oocyte Donation
FET	Frozen-thawed Embryo Transfer
eSET	elective Single Embryo Transfer
HRT	Hormone Replacement Therapy
BW	Birth Weight
LGA	Large for Gestational Age
SGA	Small for Gestational Age
OHSS	Ovarian Hyperstimulation Syndrome
HRC-FET	Hormone Replacement Cycle-FET
NC-FET	Natural Ovulatory Cycle-FET
FGR	Fetal Growth Restriction
PE	Preeclampsia
CL	Corpus Luteum
FBT	Frozen Blastocyst Transfer
CRL	Crown-Rump Length
GA	Gestational Age
GW	Gestational Week
COS	Controlled Ovarian Stimulation

PGT	Preimplantation Genetic Testing
PV	Placental Volume
uPVV	uterine Peak Venous Velocity
UtA-PI	uterine Artery Pulsatility Index
UtA-RI	uterine Artery Resistance Index
VEGF	Vascular Endothelial Growth Factor
NT	Nuchal Translucency
g DNMs	germline De Novo Mutations
HLA	Human Leucocyte Antigens
CHD	Congenital Heart Disease
ICM	Inner Cell Mass
fb-HCG	free beta-Human Chorionic Gonadotropin

ΓΕΝΙΚΟ ΜΕΡΟΣ

Introduction

As the use of assisted reproductive technology (ART) is constantly rising, the associated fetal, maternal and neonatal complications are being thoroughly investigated. The World Health Organization (WHO) predicts that infertility will be the third most serious disease worldwide of the 21st century (1).

In vitro fertilization (IVF) or IVF/Intracytoplasmic sperm injection (ICSI) pregnancies with oocyte donation (OD) are also increasing, mainly because of advanced maternal age, premature ovarian failure, reduced ovarian reserve or surrogate motherhood (2).

The conventional IVF or IVF/ICSI procedure consists of a fresh embryo transfer, followed by frozen-thawed embryo transfers (FET) in successive cycles in case of failure at the first attempt. In the “freeze-all” strategy, all embryos are thawed and transferred in subsequent cycles. A meta-analysis comparing the conventional IVF to the “freeze-all” procedure, showed that the cumulative live birth rate and ongoing pregnancy rate is similar in both methods (3).

The first successful pregnancy using FET was documented in 1983 (4). The use of FET has increased by more than 80% since 2006 thanks to the improvement in cryopreservation techniques, the increase in live birth rates, the lower risk of ectopic pregnancies, and a more physiologic hormonal and endometrial environment (5). FET concerns about half of the ART cycles in numerous countries (6), especially in cases involving OD (7, 8). Hence, it is nowadays the most common method of IVF or IVF/ICSI as it enables surplus embryos to be stored and the number of embryos per transfer to be reduced. Actually the elective single embryo transfer (eSET) is of widespread use, in order to avoid the risks associated with multiple gestations (9).

FETs are applied using various cycle regimens, that is ovulatory cycles with or without ovulation-inducing agents, or artificially induced cycles by administering oestrogen alone, followed by oestrogen in combination with progestin, which involves ovulation suppression (hormone replacement therapy (HRT) cycles) (10). Frozen-thawed embryos are transferred to either an ovulatory cycle or to a HRT cycle - usually to a HRT cycle, while a fresh embryo is transferred to an oocyte-collecting (a non-HRT) cycle (8).

These treatments can affect both genetic and maternal-uterine environmental factors related to placental development, fetal growth and birthweight (BW), (11, 12). (Epi)genetic disorders involving imprinting are found increased in IVF/ICSI because of in vitro embryo culture in different media, ovarian hyperstimulation and the consequent alteration of uterine endocrine

status (13, 14); these factors can modify gene expression patterns in follicular granulosa cells, gametes and early embryos, affecting embryo implantation, placental development and foetal growth trajectories (15, 16). Apparently, there is an impact on all three end-point offspring phenotype measures: BW, height and childhood BMI (17).

There is data on singletons following FET having an increased risk of high BW (>4.500gr) and being large-for-gestational age (LGA) (> 90th percentile), as well as a lower risk of low BW (LBW) (<2.500gr) and being small for gestational age (SGA) (<10th percentile) compared to those born after fresh ET (18-22). Concerning the “freeze-all” strategy, the incidence of maternal hypertensive disorders and LGA neonates may be elevated (3, 23), while the risk of ovarian hyperstimulation syndrome (OHSS) is decreased (8). However HRC-FET resulted in singleton BWs that were significantly lower compared to BWs following natural ovulatory cycles (NC)-FET (3, 7).

The duration of embryonic culture may be also related to neonatal BW; actually FET at the blastocyst stage (FBT) is suggested to be the most suitable method in terms of optimal fetal growth: a study showed that whereas the risk of LBW did not change in FET cycles, it decreased after a longer culturing period extended to the blastocyst stage in fresh ET cycles (24).

There is literature data associating IVF with an increased incidence of maternal and fetal complications such as hypertensive diseases, fetal growth restriction (FGR), adverse perinatal outcomes (25-27) and congenital defects (27, 28). Notably, thawed embryos are associated with a higher risk of gestational hypertension and preeclampsia (PE) (29), that may be partly attributable to the degree by which the IVF procedure impacts the maternal hormonal environment in the first trimester, when the corpus luteum (CL) is a major source of vasoactive hormones (9, 14). Therefore its presence ensures a smooth maternal cardiovascular adaptation in early gestation (14).

However the risks of antepartum hemorrhage, SGA, preterm birth, LBW and perinatal mortality are significantly lower in pregnancies following FET (30).

The findings of macrosomia and LGA after frozen blastocyst transfer (FBT) and the consequences for childhood health and risk of obesity are unclear. In vitro embryo culture might have a comparable impact on the long-term health of the offspring; since cryopreservation contains a longer culturing phase compared to fresh ET, detecting the effects of these treatments is of great importance.

As FBT conceptions present greater fetal crown-rump length (CRL) measurements during the first trimester, followed by greater BWs compared to fresh BT (18), CRL could potentially contribute to the prognosis of embryonic growth trajectories.

ΕΙΔΙΚΟ ΜΕΡΟΣ

I. Aim

The aim of this study was to compare the CRL between thawed and fresh embryos, and in subgroups according to the type of oocytes used (homologous vs donor) and the type of pregnancy (singletons vs twins), at the routine first trimester scan. In most settings, a first trimester scan is performed at 11+0-13+6 weeks of gestational age (GA) for dating, screening for preeclampsia and chromosomal abnormalities, and early detection of severe fetal defects (31, 32).

A significant difference in the measurement of CRL between thawed and fresh embryos necessitates further investigation of the potential reasons, in terms of fetomaternal factors and culture material. More longitudinal studies are encouraged in order to evaluate the impact of thawed embryos on the incidence of LGA fetuses.

II. Subjects and Methods

This was a retrospective study (July 2010 - December 2023) conducted at the Third Department of Obstetrics and Gynaecology, School of Medicine, Faculty of Health Sciences, Aristotle University of Thessaloniki, Greece. Maternal age, type of embryo transfer (thawed vs fresh, donor vs homologous oocytes), CRL, type of pregnancy (twin vs singleton gestations) were recorded.

All women had given their informed consent that their anonymized data could be used for future research. As per standard policy for audit or observational database studies not involving any intervention or modification of the management of the participants, no institutional review was required and obtained (33). All first trimester scans were performed by experienced maternal-fetal medicine specialists, according to the International Society of Ultrasound in Obstetrics and Gynaecology guidelines for first trimester scan (34).

Qualitative variables were described as n (%), and quantitative data as mean (SD). Independent Student's t-test was employed to compare the means of quantitative variables. In viable pregnancies, fetal CRL was measured in a mid-sagittal section of the embryo using transvaginal ultrasonography with care being taken to avoid inclusion of the yolk sac. CRL was expressed as z-scores of a reference population of 11-13+6 weeks (45-84mm) in fresh and frozen/thawed blastocysts after ART conceptions. CRL z-scores were calculated with the use of reference equation obtained from the general population (35).

Further subgroup analyses were conducted according to the type of oocytes used (own vs donor) and the type of pregnancy (singletons vs twins). Chi-square test (χ^2) was also

employed to investigate any association of twin gestations with thawed vs fresh embryo transfer and embryos from donor vs homologous oocytes. The level of statistical significance was set at $p=0.05$. The statistical package IBM SPSS Statistics 29.0 was used.

III. Statistical analysis

A total of 3,082 ART pregnancies including 4,044 embryos underwent first trimester scan during the study period. Of these embryos, 1,255 (40.7%) originated from thawed transfers, whereas 1,827 (59.3%) from fresh ones. The thawed embryos included 892 (71%) singleton pregnancies and 363 pairs of twins (726 embryos) (29%). The fresh embryos included 1,228 (67.2%) singleton pregnancies and 599 pairs of twins (1,198 embryos) (32.8%) (Table 1). Of note, 1,725 (56%) pregnancies were from donor oocytes and 1,357 (44%) from homologous oocytes. The donor embryo transfers included 600 (35%) twin and 1,125 (65%) singleton gestations. The homologous-oocyte embryo transfers included 362 (27%) twin and 995 (73%) singleton gestations.

Table 1. Baseline characteristics according to type of embryo transfer (thawed vs fresh) for the whole sample.

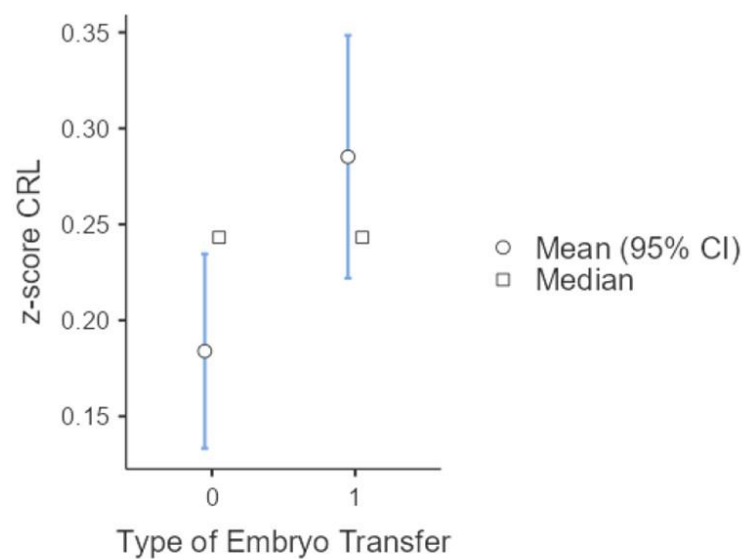
Characteristics	Thawed cycles	Fresh cycles	<i>p</i> -value	OR	95%CI
<i>Number of embryos</i>	1,618 (40%)	2,426 (60%)		N/A	
<i>Number of singleton gestations</i>	892 (71%)	1,228 (67.2%)			
<i>Number of twin gestations</i>	363 (29%)	599 (32.8%)	0.005	1.20	1.06-1.36
<i>Maternal age (mean / SD)</i>	39.8 (5.7)	35.8 (4.3)	<0.001	N/A	
<i>CRL z-score (mean/SD)</i>	0.309 (0.806)	0.199 (0.805)	<0.001	N/A	

CRL: crown-rump length, N/A: not applicable

The mean age of women who underwent thawed embryo transfer was higher compared to those following fresh embryo transfer (39.8 years vs 35.8 years, $p<0.001$). The incidence of twin gestations was significantly higher in fresh compared to thawed embryo transfer

(OR:1.20, 95% CI: 1.06-1.36, p -value= 0.629). The CRL z-score was higher in thawed compared to fresh embryos in the total sample population (0.309 vs 0.199, $p<0.001$) (Table 1, Figure 1).

Figure 1. CRL z-score according to type of embryo transfer (ET) in the whole sample (0: fresh embryo transfer, 1: thawed embryo transfer).



A subgroup analysis on twin gestations found that the mean CRL z-score was significantly higher in both thawed cycles (0.285 vs 0.184, p -value: 0.015) and in donor oocytes (0.431 vs 0.191, p -value: 0.002) (Table 2, Figure 2). An analysis on singletons also showed that the mean CRL z-score was significantly higher in thawed compared to fresh cycles (0.327 vs 0.215, p -value<0.001) (Table 3, Figure 3).

Table 2. Subgroup analysis of twin gestations

Characteristics		Thawed cycles	Fresh cycles	p -value
Maternal	age	40.1 (6.1)	35.1 (4.3)	<0.001

(mean/SD)			
CRL z-score (mean/SD)	0.285 (0.869)	0.184 (0.893)	0.015
Characteristics	Own oocytes	Donor oocytes	p-value
Maternal age (mean/SD)	37.5	43.5 (4.7)	<0.001
CRL z-score (mean/SD)	0.191 (1.05)	0.431 (0.813)	0.002

CRL: crown-rump length

Figure 2. CRL z-score, according to type of embryo transfer (ET) in twin gestations (0: own oocytes, 1: donor oocytes).

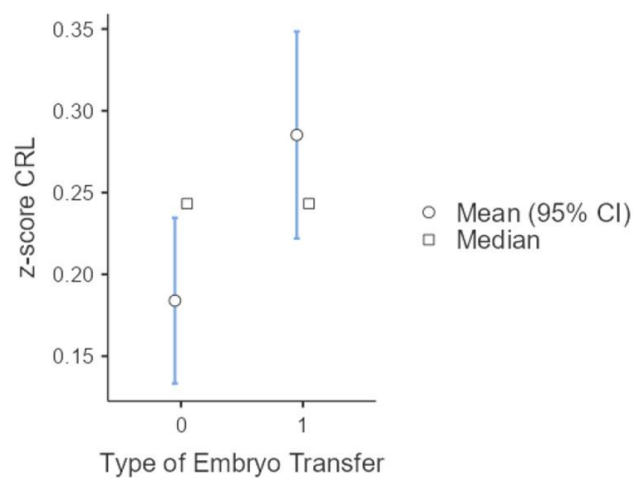
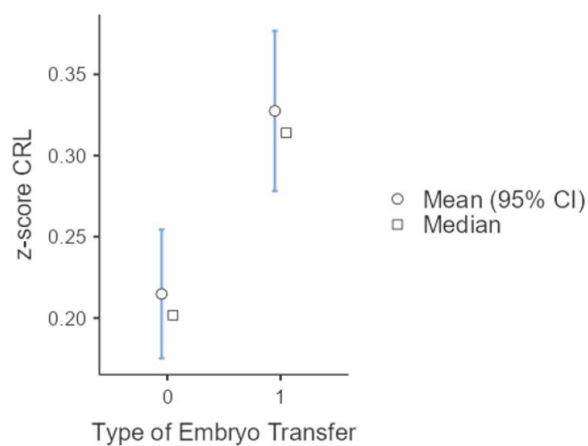


Table 3. Subgroup analysis of singleton gestations.

Characteristics	Thawed cycles	Fresh cycles	p-value
Maternal age	39.6 (5.3)	36.4 (4.3)	<0.001

(mean/SD)			
CRL z-score (mean/SD)	0.327 (0.751)	0.215 (0.709)	<0.001
Characteristics	Own oocytes	Donor oocytes	p-value
Maternal age (mean/SD)	37.6 (4.5)	43.7 (4.1)	<0.001
CRL z-score (mean/SD)	0.455 (1.17)	0.314 (0.919)	0.412

Figure 3. CRL z-scores according to type of embryo transfer in singleton gestations (0: fresh embryo 108 transfer, 1: thawed embryo transfer).



IV. Results

The main findings of this study were:

- i) the mean maternal age was significantly higher in thawed versus fresh embryo transfers in the total sample, in both singleton and twin gestations,
- ii) the mean CRL z-score at 11+0 – 13+6 GW was significantly higher in thawed compared to fresh embryos in both singleton and twin gestations,
- iii) the mean maternal age of donor oocyte cases was significantly higher than those who used their own oocytes in both singleton and twin gestations,
- iv) twin gestations were significantly higher in fresh blastocyst transfers, and
- v) the mean CRL z-score was significantly higher in twin gestations arising from donor oocytes.

V. Discussion

a) Interpretation

CRL biometry

CRL has been reported to be a predictor of BW; a GA according to CRL of 2 to 6 days lower than the expected (calculated from the day of fertilization) is associated with a higher risk of severe prematurity or low BW and catch-up growth in the first years of life (36, 37). On the contrary, a GA based on CRL of 3,5 days greater than the expected value is predictive of LGA (38, 39).

A population-based study revealed an inverse correlation between CRL measurements at the end of the first trimester and the risk of preterm birth, SGA and LBW (40). Spontaneously conceived pregnancies with increasing CRL demonstrated lower risk of stillbirth, therefore CRL measurements could potentially contribute to the prognosis of pregnancy outcome (41).

The investigation on the CRL biometry during the first trimester is consistent with greater BWs in embryos following FBT compared to fresh BT (18). There is documentation on smaller CRL measurements in fresh BT versus FBT pregnancies at 6-8 gestational weeks (GW) (42), as well as in fresh BT versus spontaneous pregnancies (43, 44). Moreover, a smaller CRL in fresh BT may be associated with SGA (18). According to a secondary analysis of a prospective cohort study, the average CRL z-score between 6 and 14 GW was notably higher in thawed compared to fresh transfers. Differences between fresh and frozen-warmed groups are more clear in early gestation and become smaller during 11-14 GW. The

likelihood of having a CRL below the 5th percentile was 68% for fresh versus 40% for thawed embryos at 6 GW, and 2% versus 1% at 14 GW, respectively (18).

Similarly, in our study, CRL measurements during the first trimester anomaly scan were significantly increased in thawed compared to fresh embryos. This difference remained significant in subgroup analyses of both singletons and twins.

Nevertheless, an explanation of the growth patterns and abnormalities in fetuses following thawed or fresh ET is complex and multifactorial. Therefore investigating any differences in CRL between thawed and fresh embryos could partially enlighten the mechanisms that lead to BW differences between these groups.

Perinatal outcomes

The available literature data suggest that obstetrical and perinatal outcomes are generally improved in FET compared to fresh ET (45), including the avoidance of endometrial asynchrony (8, 46). Better outcomes have been particularly reported in NC-FET (30, 47, 48), as a result of a more natural intrauterine environment allowing physiological early placentation. On the contrary, controlled ovarian stimulation (COS) in fresh ET may alter endometrial angiogenesis and receptivity (47). In other studies the rates of implantation, pregnancy, and live birth after HRC-FET are comparable to those after NC-FET (49). The differences in BW between the two groups become apparent and statistically significant in the last trimester of pregnancy from GW 33 onwards. Concerning the risk of LGA, the differences appear later starting from GW 36, becoming significant slightly earlier for boys than girls (50).

A meta-analysis of 11 studies and two large cohort studies found a reduced risk of SGA and FGR by about 45% and an increased risk of LGA by 30-90% in FBT compared to fresh ET pregnancies (30, 51, 52). In another study there was a 3-fold higher rate of SGA in the fresh BT group compared to the FBT group (53). A retrospective analysis of 5,406 embryos also revealed that BWs of neonates from thawed was higher than those arising from fresh embryos (54). There were no differences in Apgar scores (20), no difference in the risk of perinatal mortality (55) and no long-term weight and height differences above 5 years of age (56). There are reports suggesting an altered sex ratio toward more males following single FBT (29). Also, fresh ET results in lower BW infants compared to those of natural conception (20).

Moreover, singletons conceived through HRC-FET are at higher risk of post-term delivery and caesarean section than those conceived through NC-FET; this is to be taken under

consideration as late-term and post-term pregnancies present a higher incidence of perinatal mortality and morbidity other than macrosomia (57).

Three systematic reviews and meta-analyses on perinatal outcomes after blastocyst versus cleavage stage embryo transfer that included more than 100,000 singletons, found a significantly higher rate of preterm birth (<37 weeks of gestation) and very preterm birth (<32 weeks of gestation) after blastocyst compared to cleavage stage transfer (58).

Aging-related infertility

As higher maternal age is related to more obstetric and maternal complications, it is reasonable to proceed with thawed embryo transfer (45). A retrospective cohort study confirmed that maternal age is higher in thawed compared to fresh embryo transfers (59). The same finding was supported by other published data (20), which were in agreement with the results of our study. Also, maternal age in case of OD was significantly increased compared to homologous gestations, which is reasonably explained; in 2014, up to 12% of all IVF cycles in the U.S. were performed using donor oocytes with a high pregnancy rate from women in their 50s being observed (>35%) (60). OD currently represents the main option of aging-related infertility (60).

Twin gestations

Twin gestations are still a relatively common occurrence in ART (61). The results of our study indicated a higher prevalence of twin gestations in fresh compared to thawed embryo transfer; this could be explained by the number of embryos transferred in each case.

Notably in our study, the mean CRL z-score was significantly higher in cases of twin gestations arising from donor oocytes, which was not observed in singletons. One retrospective cohort study found that the median CRL was significantly higher in cases of embryos following OD compared to those from homologous oocytes, in a sample which included both singleton and multiple gestations (62).

Furthermore, a comparative study on cost-effectiveness of single embryo transfer followed by an additional thawed embryo transfer, compared to double embryo transfer, showed that in women above 32 years of age the double embryo transfer was more effective in fresh cycles (63), explaining the higher prevalence of twins in fresh embryo transfer. Increasing the success rate in the area of ART is crucial, taking into consideration the high cost of the procedure (64). A meta-analysis of individual patient data from randomized trials showed that eSET is related to a higher chance of delivering a full-term, healthy neonate compared to double embryo transfer, yet it is also associated with lower pregnancy rate in fresh cycles

(23). Despite the fact that this difference is completely overcome by an additional thawed embryo transfer, there is still a tendency towards maximizing the chances of live birth at the first attempt and not cumulatively (23).

Elective Single Embryo Transfer

The eSET approach is the only effective way of minimizing the risk of multiple gestations and can also be applied to women 36-39 years old (65). Preimplantation genetic testing for aneuploidies (PGT) could also enhance the strategy of eSET, without compromising the results of ART. Although evidence suggests that PGT improves live birth rates in women above 35, the Greek legislation does not allow its implementation in women below 40 years of age without any other indication, thus discouraging the eSET practice (66).

b) Impact of IVF on placental formation

Implantation initiates placental development by triggering modifications in the decidua (the maternal part of the placenta) and spiral arteries (67). The decidua regulates trophoblastic invasion into the uterine wall, inhibiting trophoblast cells from penetrating too deep beyond the decidual layer into the myometrium (placenta accreta) (10). Otherwise, a shallow invasion of the trophoblast cells results in PE (68).

Endometrial thickness is related to ART procedures (69). There is documentation on epigenetic gene expression changes in the endometrium and in placentas following HRC-FET, NC-FET and fresh ET (70, 71). Apparently the effect of IVF regimens on DNA methylation in the placenta is genotype-specific (72). There might be differences in placental morphology, volume and placental functions among the IVF cycle regimens, and these differences can have an impact on BW (69) (10).

FET vs fresh ET vs natural conception

Studies have also demonstrated that first-trimester placental volume (PV) after FET is increased compared to fresh ET pregnancies (17, 73). Apparently endometrial receptivity is inferior in fresh ET gestations compared to FET and to naturally conceived groups (25, 30, 47, 74). Irrespective of the underlying cause, reduced PV in fresh ET pregnancies predisposes to higher risk of PE (17).

A study showed that placental developmental parameters (PV, uterine peak venous velocity (uPVV), uterine artery pulsatility index (UtA-PI) and uterine artery resistance index (UtA-RI)) presented no significant differences among FET, fresh ET and natural conception pregnancies in the first trimester (75). From the second trimester onwards, FBT pregnancies

had a lower UtA-PI and UtA-RI compared to the fresh ET group, which suggests a lower distal vascular resistance, with better fetal growth and a lower risk of SGA (75). In particular, UtA-PI was 14% lower and BW centile was 7% higher in FET pregnancies compared to fresh (17, 75). These indices were comparable between fresh ET and natural conception pregnancies (73).

Increased UtA-PI is associated with abnormalities in trophoblast invasion and is predictive of placenta-related pregnancy complications such as PE, hypertensive disorders and FGR (76-79); it can be already elevated at 11 GW in pregnancies that develop PE (80, 81). There is a significant inverse correlation between UtA-PI and blood flow volume in the uterine arteries (UtAs) that reinforces the concept of reduced blood perfusion in placental dysfunction (82).

HRC-FET vs NC-FET vs natural conception

The placental basal plate is modified in HRC-FET compared to NC-FET following the endometrial preparation for embryo transfer (10, 83). Gonadotropin administration during implantation in HRC-FET affects the production of vasculogenic and angiogenic factors, affecting trophoblast migration and vascular invasion in the endometrium (30, 47, 84). The Rohr fibrinoid layer between the villous structure and the extravillous trophoblast layer is thickened in HRC-FET, while the decidual layer is underdeveloped (68), resulting in abnormal placentation related to hypertensive disorders and FGR (68, 85-87).

However an important underdevelopment of the decidua might allow trophoblast cells to proliferate and invade to a higher degree. This could explain certain findings of higher BWs in the HRC-FET group compared to NC-FET group (10).

Another study reported that HRC-FET was associated with higher rates of PE (12,8% versus 3.9%; $p= 0.02$) and PE with severe features (9.6% versus 0.8%; $p= 0.002$) compared to NC-FET (88). Also, a randomised trial where 75% of the FETs were performed in a natural, ovulatory cycle, found no significant risk of PE and no differences in the first-trimester placental developmental parameters compared to those of fresh ET (89). However there are cases of fresh ET pregnancies with low PV that develop PE despite normal uterine UtA-PI values (17, 90-92).

Cryopreservation techniques

Cryopreservation techniques might also affect placental development. (93, 94). In several animal models ART regimens led to numerous placental abnormalities, such as impaired placental steroid metabolism and a larger placental/fetal weight ratio in mice (95, 96), and

alterations in placentome morphology and blood vessels in cattle (97). In humans there is an increased risk of placental abnormalities, such as placenta praevia (98).

Oocyte donation gestations

Concerning OD pregnancies, in most cases they involve the use of both oocyte and embryonic cryopreservation and present a higher incidence of SGA, LBW and histological decidual-placental abnormalities compared to homologous IVF gestations (99). Normally during implantation, trophoblastic cells expressing human leucocyte antigens (HLA)-C regulate decidual neovascularization and modulate the spiral arteries. In case of OD the trophoblastic HLA-C is completely allogenic and may alter the functionality of uterine natural killer (NK) cells. As a result, maternal blood supply to the placenta is affected, triggering the endothelial cascade to PE and FGR (100) (101).

c) Impact of IVF on embryonic and fetal evolution

While spontaneous pregnancies usually develop in the presence of one CL, IVF involves two extremes—either formation of supra-physiologic numbers of CLs associated with ovarian stimulation in fresh IVF cycles, or hypothalamic-pituitary suppression and absence of the CL in artificial, programmed cycles routinely used for donor egg recipients and FET (14).

CL secretes vasoactive agents such as relaxin, essential in maternal systematic vasodilation (102). Absence of CL is associated with reduced aortic compliance and increased risk of PE, demonstrating the critical role of CL in the physiologic transformation of the maternal cardiovascular system in early gestation (14, 103).

Hormone-Replacement Cycle (HRC)-FET

In HRC-FET the hypothalamic-pituitary-ovarian axis is suppressed by consecutive administration of oestrogens and progesterone to prepare the endometrium for transfer (102). Oestrogen concentration regulates vascular development both in the endometrium and in the trophoblast through angiogenic growth factors (104). In animal models, artificially elevated circulating oestrogen levels during the first trimester stimulate the production of vascular endothelial growth factor (VEGF) (105); high VEGF levels affect trophoblastic vascular remodeling with a measurable negative impact on uterine and umbilical hemodynamics, observed at Doppler analysis (106).

As endometrial luteinization is affected the implantation window is disturbed (107, 108), as shown in histological endometrial changes and expression of implantation window markers (109, 110). Although estradiol and progesterone are replaced during the first trimester in

programmed FET cycles, vasoactive products of the CL such as relaxin are not administered (111). The absence of CL and the increased trophoblastic invasiveness in HRC-FET predisposes to a higher risk of hypertensive disorders in pregnancy and neonatal macrosomia (BW>4500gr) compared to NC-FET (112) (25, 113).

Regarding BWs after luteal phase support, administration of oestrogen combined with progesterone led to increased neonatal weights compared to pregnancies without luteal phase support or those who underwent luteal phase support without oestrogen (10, 114).

Besides pregnancy-induced hypertension, HRC-FET has been especially associated with an increased amount of bleeding during delivery (83) and a higher incidence of placenta accreta (29). The underlying pathophysiology could be related to the endometrial injury following embryo transfer or the altered epigenetic status due to the in vitro embryo culture and cryopreservation (51, 115, 116), as well as to the hormonal disturbances and patients' infertility itself (117, 118).

Fresh ET

Concerning fresh ET, high blood concentrations of oestradiol and progesterone following controlled ovarian stimulation can last at least through the first trimester (119) and may create a suboptimal environment for implantation and placentation, indicated by altered steroid receptor expression (109, 110). This could justify the differences observed in first-trimester placental developmental parameters between fresh ET and naturally conceived pregnancies (120). Also, it predisposes to FGR (113, 117) and PE (17, 47), explaining the increased incidence of LBW and SGA compared to natural conception or HRC-FET pregnancies (113, 117).

Natural Cycle (NC)-FET

Unlike fresh ET, NC-FET seems to provide a more physiological uterine environment as the CL is present, favorising a normal early placentation and embryonic development (17, 119). There is no significant difference in the frequency of preeclampsia between modified NC-FET and spontaneous conception (14).

Blastocysts versus cleavage-stage frozen embryos

A study showed no apparent difference in BW between blastocysts and cleavage-stage frozen embryos, suggesting that the manipulation of the endometrium could affect BW regardless of the culturing period of the embryo (10). However there is documentation on a decreased risk

of LBW in fresh ET cycles following a culturing period extended to the blastocyst stage (24). Other studies also revealed a significantly lower risk for SGA infants after FBT (121, 122). A possible explanation is that cryopreservation techniques may cause epigenetic modifications at the early embryonic stages, affecting DNA methylation related to the growth potential of the fetus (55, 123, 124). The overgrowth observed in FBT fetuses may be as well a selection process with higher quality embryos surviving the prolonged freezing and thawing procedure (55).

Frozen-thawed blastocysts are generally exposed to higher risks compared to those transferred on fresh cycles or to frozen-thawed embryos transferred at cleavage-stage on day 2-3. In fact, longer in vitro embryonic growth in association with freezing-thawing procedures with the use of cryoprotectants, may cause a biological damage to the cells (125, 126). The significantly higher incidence of preterm birth (<37 GW) and very preterm birth (<32 GW) after blastocyst compared to cleavage stage transfer is related to genetic and epigenetic changes occurring in the trophoctoderm cells that alter placentation and implantation (58).

CRL

CRL of IVF embryos at the end of the first trimester is 2-3 days in advance compared to embryos conceived in vivo (127, 128). As the development of human embryos during the first weeks is believed to be homogeneous, this difference might be due to an accelerated in vitro phase (92).

Several factors could contribute to the observed differences in CRL growth, such as changes in the condition of the endometrium (10, 52), epigenetic modifications in the trophoblast (129), variations in ovarian hormone levels during the peri-implantation phase (88), embryo quality, differences in blood flow to the uterus and disparities in cardiovascular health among the groups studied (103). Moreover, the practice of culturing the frozen embryos and transferring them on day 2 of the luteal phase means that many will have undergone an additional cleavage and be more advanced in development than the fresh embryos transferred on day 2 (19).

Fetal CRL appears positively associated with higher maternal age, parity>1, folic acid use, negatively associated with smoking and not associated with maternal anthropometrics (40). Ovulation induction in infertile women was related to a reduction in CRL during the first trimester and increased risk of low BW compared to fertile women (130). There is evidence to suggest that pregnancies on the ground of infertility achieved with or without medical

assistance, lead to smaller infants compared to those of fertile women (10, 120). Histological studies showed that the endometrium of infertile women even in ovulatory menstrual cycles is often unfavorable for implantation, that may be due to subtle ovarian dysfunction. In case of achieved pregnancy, these endometrial abnormalities causally related to infertility may influence fetal development (131).

Nuchal translucency

Interestingly, nuchal translucency (NT) also demonstrated an increase in euploid fetuses following BT with or without cryopreservation compared to embryos conceived spontaneously. Median NT thickness was about 15% higher in the fresh BT group and 25% higher in the FBT group than in controls. No difference in NT measurements was found between FBT and fresh BT pregnancies. The obstetric outcome was normal (132).

The classic underlying pathophysiology for increased NT thickness in euploid fetuses is related to abnormalities of the heart and great arteries, cardiac dysfunction, altered composition of the extracellular matrix of the nuchal skin, venous congestion in the head and neck, abnormal or delayed development of the lymphatic system, impaired fetal movements, fetal anemia or hypoproteinemia, congenital infections and abnormal migration of neural crest cells (133, 134). In a study of ART fetuses, there was a progressive increase in NT from the control group to the fresh BT group, and from the latter to the FBT group, which presented the highest median dNT (132). This tends to exclude ovarian hyper-stimulation as the main cause. In contrast, the increased NT measurement in both IVF fresh and frozen groups could be related to the duration of the in vitro cell growth, which is about 2 days longer in blastocyst transfer compared to transfer of cleavage-stage embryos. The observed highest median dNT in the frozen group may also be related to the use of more invasive techniques for blastocyst cryopreservation (such as assisted hatching and assisted shrinkage). These procedures, in addition to the use of high concentrations of potentially toxic cryoprotectants, may cause a subclinical damage to the blastocysts that provokes an inflammatory reaction, presenting as increased NT at a later stage (125, 135).

Also, free beta-human chorionic gonadotropin (b-hCG) showed a 30% increase in FBT and a 10% increase in fresh BT compared to controls (132). Other studies have also associated ovarian stimulation and ART conceptions with higher b-hCG, as well as lower pregnancy-associated plasma protein A (PAPP-A) levels (136, 137).

Therefore, first trimester combined testing at 9+0 – 13+6 GW for common fetal trisomies can be altered by ART procedures (132). Apparently the screening test variables should be

adjusted according to the mode of conception (natural or assisted) together with the other factors that can have an impact on the accuracy of the algorithm (maternal weight, ethnicity, smoking, diabetes, multiple pregnancy, vanishing embryos) (132).

Impact of IVF on offspring phenotype and long-term health

Ovarian stimulation regimens following different ART methods alter nuclear and cytoplasmic maturation of the oocyte as well as epigenetic imprinting (138, 139). Changes in DNA methylation patterns may influence fetal growth kinetics (114, 140). Also, the indirect actions of these regimens on endometrial and luminal characteristics may impact not only BW but growth trajectory ex utero as well (2, 141). The presence of GnRH receptors in cumulus-oocyte complexes and preimplantation embryos (142), as well as in the endometrium (143) provides a potential mechanism by which ovarian stimulation can directly impact offspring phenotype (144). Genes that acquire their imprints later in oocyte development are the most susceptible to perturbations (91). In animal models such epigenetic modifications are responsible for the increased risk of LGA in FET gestations (145); in cattle the large offspring syndrome occurs significantly more often after IVF/ART procedures (146). There is data on the large offspring syndrome being genetically similar to the Beckwith-Wiedemann's syndrome in humans, which is indeed epigenetic (147).

Even though BW differences related to ART procedures become statistically and clinically insignificant after the age of 5 years (56), according to the developmental origins of health and disease (DOHaD) theory both low and high BWs are related to high cardiometabolic risk profiles and obesity in adulthood (148, 149) (150); LBW (<2500gr) has been associated with higher risk of adult-onset disease including increased insulin resistance (151), type II diabetes (152), kidney disease (153), and cardiovascular diseases (154). On the other hand macrosomia (BW>4000gr) or LGA (>90th percentile) is also associated with increased morbidity in adulthood including chronic kidney disease (153), obesity and adverse arterial profiles (155).

Birth defects

As ovarian hyperstimulation impacts oocyte quality, it contributes to the origins of birth defects and developmental deficiencies that manifest at birth or at later stages of life (156, 157). A systematic review of 45 cohort studies showed that ART neonates have a higher risk of birth defects compared to those naturally conceived (27). An increased incidence of congenital malformations in infants born after cryopreservation compared to fresh ET is also described (158, 159). Belva et al (2008) reported higher rates of major malformations in

cryopreserved embryos following ICSI (6.4%) compared to cryopreserved embryos without ICSI (3.1%) and fresh embryos with ICSI (3.4%) (159).

It is documented that the origins of birth defects and possibly other childhood and adult diseases can arise during the earliest stages of gonad development and gamete differentiation and are almost exclusively related to the oocyte (160). This was stated even since the 17th century by William Harvey on Human Reproduction: “*Ex ovo omnia*”: everything comes from the egg (161). Physiological studies demonstrated that a network of cellular interactions mediate critical aspects of oocyte metabolism that have a direct effect on post-fertilization development of the conceptus (162, 163).

As epigenetic marks are erased in the primordial germ cells, epigenetic modifications can appear before and after fertilization (164). Transitions between open and closed chromatin states during oogenesis and early embryogenesis boost genome-wide epigenetic reprogramming that involve DNA methylation or histone changes in phosphorylation and methylation (160, 165). Epimutation refers to an aberrant DNA methylation or histone modification pattern. Most imprinting defects are primary epimutations that happen in the absence of DNA mutations (166). Differentially methylated regions in imprinted genes are protected from demethylation that occurs in the early embryo during preimplantation development (164).

Germline de novo mutations (gDNMs) are new mutations occurring within one generation; they provide offspring with additional genetic variants and account for nearly 90% of total DNMs (167)(168). There is data showing that DNMs can accumulate with DNA replication during spermatogonium division or the double-strand break repair process of oocytes (167).

Specific ART procedures, such as ovarian stimulation and trigger protocols, contribute to the number of maternal DNMs; both processes were found to stimulate the oocyte meiosis resumption (169) and maturation (170). Another study also revealed that meiotic recombination could play a role in the formation of maternal DNMs (171).

Changes in DNA methylation and/or transcription of some transposable elements and imprinted genes were found in placenta samples and in cord blood following ART (70). A Finnish study on human placentas correlating DNA methylation changes with phenotypic findings, suggested that an intronic rs10732516 polymorphism in the sixth binding site of CCCTC-binding factor in the *H19* imprinting control region, is related to the effects of ART in a parent-of-origin manner (72).

Human oocytes and embryos often fail to stop the propagation of aneuploid cells but maintain their ability to repair DNA damages including those introduced by the fertilizing sperm (160).

There is a 29% increased risk of birth defects in newborns spontaneously conceived by subfertile couples and the risk is further increased (34%) when conception is achieved by ART (160). Genetic factors in infertile couples in addition to the variables involved in ART (hormones, culture media, prolonged embryo culture, cryopreservation) may have adverse epigenetic effects (172).

The application of ART was associated with a risk of congenital heart diseases (CHD) through accumulating functional gDNMs. However, although the non-coding functional mutations are important mediators between ART and CHD, maternal DNMs accounted for only a small proportion of these mutations and the majority were originated from the fathers. Thus, ART procedures may not directly contribute to the risk of CHD (168).

Embryo quality

There is growing evidence that the human endometrium serves as a biosensor of embryo quality (173). In vitro studies showed that decidualized endometrial stromal cells have the capacity to discern high- and low-quality embryos and selectively respond to developmentally impaired embryos by inhibiting the secretion of certain growth factors and cytokines (174). The total methylation levels of high-quality blastocysts are similar, whereas low-quality blastocysts present different and variable methylation levels (175). Good quality embryos present an increased implantation rate (10). Blastocysts with more advanced inner cell mass (ICM) become heavier infants, compared to those with a less advanced ICM. In contrast, the trophectoderm grade was not related to BW (176).

Smaller (16-18mm diameter) follicles are associated with a decrease in maturation and fertilization rates and the production of poorer quality embryos (177). Such embryos, either fresh or thawed, are related to increased child height and lower BW through epigenetic modifications (91). Developmentally compromised embryos appear to be metabolically more active in terms of amino acid turnover, producing higher levels of reactive oxygen species and ATP compared to competent embryos (178). This altered homeostasis and disturbed metabolism may alter normal growth in the long-term (179).

Thus, there is documentation that the phenotypic plasticity of the oocyte and embryo seems to be greater when they are less mature and of poorer quality, and follicle diameter at the ovulation trigger might be a potential predictor for offspring phenotype (91).

Peri-conceptual undernutrition

Factors in the peri-conceptual period such as undernutrition, can affect fetal growth with negative impact for long-term health of the offspring, even if BW is normal (180). A typical example is that individuals conceived during the Dutch famine of World War II presented elevated incidence of coronary artery disease, a more atherogenic lipid profile, obesity, pathologic blood coagulation markers and higher responsiveness to stress compared to unexposed individuals or those exposed during mid- or late-pregnancy, even with BWs in the normal range (181). These observations were confirmed by animal studies: peri-conceptual undernutrition in sheep can alter fetal growth (182). In rats, a low-protein diet to the mother only during preimplantation period resulted in lower cell numbers in the inner cell mass and trophectoderm, altered expression of growth-regulating imprinted genes, reduced BW and hypertension in the offspring (183, 184). In mice, maternal low-protein diet exclusively during preimplantation did not affect fetal growth but was related to neonatal hypertension and excessive growth (185).

There is also data on gametes or even preimplantation embryos adapting to environmental alterations in response to paternal diet, with long-term health consequences for the offspring (186, 187).

Aspects of in vitro embryo culture

Several studies relating neonatal and childhood characteristics with IVF methodology have focused on aspects of in vitro culture, including the impact of culture media (188), length of embryo culture (189) and embryo freezing (19).

The embryo culture medium affects embryo quality, the ability of preimplantation embryos to properly implant and pregnancy chances (190). Culture media comparison studies have observed differences in early intrauterine growth (191), placental weight (191), birth outcomes including BW (192), and growth during the first 2 years of life (193). Certain culture conditions are associated with FGR in early to mid-pregnancy, succeeded by substantial increases in placental size and accelerated fetal growth toward the end of gestation (194).

The effects of culture media on neonatal BW could be attributed to a direct effect on the epigenome of the developing preimplantation embryo (195); studies showed culture medium-dependent differences in mRNA expression levels of genes involved in apoptosis, protein degradation, metabolism and cell-cycle regulation (196). Animal studies have verified that culture medium can affect the epigenetic regulation of imprinted genes mainly in the placental tissue, and to a lesser extent in the embryo (197). Via this mechanism Vitrolife G1.3

medium might be related to increased fb-hCG levels, increased number of trophoblast cells and fetal growth (195). Other studies have showed that the use of Vitrolife G5 and human tubal fluid (HTF) resulted in different gene expression profiles in human preimplantation embryos at the blastocyst stage (198) and confirmed an association between culture media and BW (190). A comparative study between singletons after embryo culture in Vitrolife versus Cook media also concluded that the Vitrolife group presented significantly higher circulating levels of fb-hCG at 12 GW and a more advanced GA (199).

HCG is a glycoprotein produced by the syncytiotrophoblast cells that is crucial to embryonic development by promoting progesterone production, angiogenesis in uterine vasculature, uterine growth, differentiation of growing cytotrophoblast cells, quiescence of myometrial contractions and is also implicated in development and growth of fetal organs (200). Low first-trimester fb-hCG levels are associated with gestational hypertension, gestational diabetes, FGR and SGA (199). The higher fb-hCG levels in the Vitrolife group is consistent with the increased fetal growth (199).

Orasanu et al (2006) investigated the effect of four different IVF culture media on serum hCG concentrations measured on Day 15 after embryo transfer in singleton viable pregnancies and concluded that Vitrolife G1.3 medium was associated with the highest serum hCG levels compared to the other media (201).

Enrichment of IVF culture media with the appropriate nutrients is vital for fetal development and long-term health (199). In human and animal models the addition of amino acids to the culture medium improved *in vitro* embryo development (202). A meta-analysis indicated that the use of hyaluronan-enriched culture media resulted in increased implantation and live birth rates (203). Other studies have also suggested that the protein source in the culture media influences live birth rate (204) and BW (189).

Concerning culture time, blastocysts necessitate a more extended in-vitro culture of 5 to 6 days compared to cleavage stage embryos (2 to 4 days), that exceeds the timing of embryonic genomic activation, potentially affecting the embryo and offspring. The culture media used for extended culture periods differ from those used for growth at cleavage stage (205).

It has been suggested that the extended embryo culture time might be associated with higher BWs (206) and a doubling in the number of LGA infants (160). In fresh ET cycles the extended culture time of blastocysts is known to decrease the risk of low BW (24).

Male sex was found to be more affected by extended embryo culture times concerning fetal growth; in a single-centre study of 1295 singletons, term boys born after HRC-FBT presented a significantly greater risk of LGA compared to girls (207). In a nationwide cohort study from the US consisting of 180184 singletons (208), FET had a slightly stronger association with larger birthweights of term boys compared to girls. According to certain studies, there was no increased risk of birth defects after extended time in culture (58, 122).

Concerning cryopreservation techniques, vitrification is an ultrarapid method that has prevailed slow freezing. There is data on a slightly favorable cryosurvival and clinical pregnancy rate for vitrification compared to slow freezing, although perinatal outcomes are similar between the two methods (93, 94). However, the high concentrations of cryoprotectants required in vitrification to avoid ice crystal formation have raised concerns about possible health risks for the offspring (209).

d) Conclusions

The differences in CRL observed at the different ART techniques suggest that the type of hormones and protocols used could have different effects on trophoblastic invasiveness, that are clinically visible from the second trimester. Considering the correlation between CRL and BW, CRL measurements could be useful as a prognostic tool of fetal growth potential.

The major strength of our study is its sample, which provides a significant statistical power regarding the results. To date, this is the only study providing data on the differences between CRL in thawed and fresh embryos during the nuchal translucency scan. The main limitation of the study is its retrospective nature and other maternal co-factors that could play a role to the findings, such as comorbidities; cardiovascular disease, including hypertension, body mass index, smoking. Of note, the risk of bias is eliminated because of the electronic recording of the data.

As epigenetics plays a major role in placental morphology and fetal growth, further studies are needed to assess changes in placental function and epigenetic modifications induced by each of the ART techniques. These novel findings also highlight the importance of taking the mode of conception into account in fetal growth follow-up, and they may lead to the development of research protocols (preventive or therapeutic) to minimize the adverse fetal outcomes potentially linked to ART procedures.

The fact that human embryos cultured in vitro are sensitive to their environment should lead to increased awareness, mechanistic studies and legislative adaptations to protect IVF offspring during the first few days of their existence.

ΠΕΡΙΛΗΨΗ

Εισαγωγή: Νεογνά που προέρχονται από μεταφορά κατεψυγμένων εμβρύων παρουσιάζουν σημαντικά μεγαλύτερο βάρος γέννησης σε σύγκριση με νεογνά μετά από άμεση μεταφορά εμβρύων (φρέσκων). **Σκοπός:** Σύγκριση του κεφαλουραίου μήκους (CRL) μεταξύ κατεψυγμένων και φρέσκων εμβρύων και προσέγγιση της παθοφυσιολογικής ερμηνείας των διαφορών. **Υλικό και μέθοδοι:** Πρόκειται για μία αναδρομική μελέτη (Ιούλιος 2012 – Δεκέμβριος 2023) που πραγματοποιήθηκε στην Τρίτη Πανεπιστημιακή Μαιευτική και Γυναικολογική κλινική της Ιατρικής Σχολής του Αριστοτελείου Πανεπιστημίου Θεσσαλονίκης. Όλες οι κυήσεις υποβοηθούμενης αναπαραγωγής υποβλήθηκαν σε υπερηχογραφικό έλεγχο ρουτίνας μεταξύ 11+0ης – 13+6ης εβδομάδας κύησης. Καταγράφηκαν και αναλύθηκαν οι εξής παράμετροι: ηλικία μητέρας, τύπος εμβρυομεταφοράς (κατεψυγμένα ή φρέσκα), ωάρια δότριας σε σχέση με αυτόλογα ωάρια, κεφαλουραίο μήκος, μονήρεις και δίδυμες κυήσεις. **Αποτελέσματα :** Η μέση μητρική ηλικία ήταν σημαντικά μεγαλύτερη στις κυήσεις κατεψυγμένων εμβρύων σε σύγκριση με φρέσκων

(39.8 vs 35.8 χρόνια, p -value<0.001). Το μέσο CRL z-score διαπιστώθηκε σημαντικά υψηλότερο στα κατεψυγμένα σε σχέση με τα φρέσκα έμβρυα (0.309 vs 0.199, p -value<0.001). Από την ανάλυση της υποομάδας των μονήρων κυήσεων προέκυψε πως το μέσο CRL z-score ήταν υψηλότερο στα έμβρυα κατεψυγμένων βλαστοκύστεων σε σχέση με φρέσκων (0.327 vs 0.215, p -value<0.001). Αντίστοιχα, η ανάλυση των διδύμων κυήσεων κατέδειξε πως το μέσο CRL z-score ήταν υψηλότερο στις κατεψυγμένες βλαστοκύστες σε σχέση με φρέσκες (0.285 vs 0.184, p -value: 0.015) και στις λήπτριες ξένων ωαρίων σε σχέση με αυτόλογων (0.431 vs 0.191, p -value: 0.002). **Συμπεράσματα :** Οι διαφορές στις μετρήσεις CRL μεταξύ κατεψυγμένων και φρέσκων εμβρύων ενδεχομένως αποτελεί μια πρόιμη ένδειξη της επακόλουθης διαφοράς στην υπερηχογραφική εκτίμηση του εμβρυικού βάρους και στο βάρος γέννησης. Αυτά τα ευρήματα υπογραμμίζουν την ανάγκη για περαιτέρω έρευνα όσον αφορά στα υποκείμενα αίτια, συμπεριλαμβανομένων των μητρικών παραγόντων, των διαφόρων πρωτοκόλλων και θεραπευτικών καλλιεργητικών υλικών που χρησιμοποιούνται στις τεχνικές εξωσωματικής γονιμοποίησης.

Λέξεις-κλειδιά: Υποβοηθούμενη αναπαραγωγή, κατεψυγμένα, φρέσκα, κεφαλουραίο μήκος, μέτρηση.

SUMMARY

Background and Objectives: Neonates born from thawed embryo transfers tend to have significantly higher birthweight compared to those following fresh embryo transfers. **Aim:** To compare the crown-rump length (CRL) between thawed and fresh embryos and to approach the underlying pathophysiology that could explain the differences. **Subjects and methods:** This was a retrospective study (July 2010-December 2023) conducted at the Third Department of Obstetrics and Gynecology, School of Medicine, Faculty of Health Sciences, Aristotle University of Thessaloniki, Greece. All assisted reproductive technology (ART) pregnancies underwent a routine scan at 11+0 – 13+6 gestational weeks. Maternal age, type of embryo transfer (thawed vs fresh, donor vs their own oocytes), CRL, twin and singleton gestations were analyzed. **Results:** The mean maternal age in thawed was significantly higher than in fresh embryos (39.8 vs 35.8 years, p -value<0.001). The mean CRL z-score was significantly higher in thawed compared to fresh embryo transfers (0.309 vs 0.199, p -value<0.001). A subgroup analysis on singleton gestations showed that the mean CRL z-score was higher in thawed blastocysts compared to fresh (0.327 vs 0.215, p -value<0.001). Accordingly, an analysis on twins revealed that the mean CRL z- score was higher in thawed blastocysts (0.285 vs 0.184, p -value: 0.015) and in oocytes' recipients compared to own oocytes' cases (0.431 vs 0.191, p -value: 0.002). **Conclusion:** The difference in CRL measurements between thawed and fresh embryos may be a first indication of the subsequent difference in sonographically estimated fetal weight and birthweight. This finding highlights the need for additional research into the underlying causes, including maternal factors, IVF protocols and the culture media used.

Keywords: ART; thawed; fresh; embryo; CRL; measurement.

REFERENCES

1. Kushnir VA, Barad DH, Albertini DF, Darmon SK, Gleicher N. Systematic review of worldwide trends in assisted reproductive technology 2004-2013. *Reprod Biol Endocrinol*. 2017;15(1):6.
2. Pennings G, de Mouzon J, Shenfield F, Ferraretti AP, Mardesic T, Ruiz A, et al. Socio-demographic and fertility-related characteristics and motivations of oocyte donors in eleven European countries. *Hum Reprod*. 2014;29(5):1076-89.
3. Zaat T, Zagers M, Mol F, Goddijn M, van Wely M, Mastenbroek S. Fresh versus frozen embryo transfers in assisted reproduction. *Cochrane Database Syst Rev*. 2021;2(2):Cd011184.
4. Zeilmaker GH, Alberda AT, van Gent I, Rijkmans CM, Drogendijk AC. Two pregnancies following transfer of intact frozen-thawed embryos. *Fertil Steril*. 1984;42(2):293-6.
5. Luke B, Brown MB, Wantman E, Forestieri NE, Browne ML, Fisher SC, et al. The risk of birth defects with conception by ART. *Hum Reprod*. 2021;36(1):116-29.
6. De Geyter C, Calhaz-Jorge C, Kupka MS, Wyns C, Mocanu E, Motrenko T, et al. ART in Europe, 2014: results generated from European registries by ESHRE: The European IVF-monitoring Consortium (EIM) for the European Society of Human Reproduction and Embryology (ESHRE). *Hum Reprod*. 2018;33(9):1586-601.
7. Kawwass JF, Monsour M, Crawford S, Kissin DM, Session DR, Kulkarni AD, et al. Trends and outcomes for donor oocyte cycles in the United States, 2000-2010. *Jama*. 2013;310(22):2426-34.
8. Wong KM, van Wely M, Mol F, Repping S, Mastenbroek S. Fresh versus frozen embryo transfers in assisted reproduction. *Cochrane Database Syst Rev*. 2017;3(3):Cd011184.
9. Henningsen AA, Gissler M, Skjaerven R, Bergh C, Tiitinen A, Romundstad LB, et al. Trends in perinatal health after assisted reproduction: a Nordic study from the CoNARTaS group. *Hum Reprod*. 2015;30(3):710-6.
10. Ishii R, Shoda A, Kubo M, Okazaki S, Suzuki M, Okawa R, et al. Identifying a possible factor for the increased newborn size in singleton pregnancies after assisted reproductive technology using cryopreserved embryos, in comparison with fresh embryos. *Reprod Med Biol*. 2018;17(3):307-14.
11. Pritchard CW, Sutherland HW, Carr-Hill RA. Birthweight and paternal height. *Br J Obstet Gynaecol*. 1983;90(2):156-61.
12. Little RE, Sing CF. Genetic and environmental influences on human birth weight. *Am J Hum Genet*. 1987;40(6):512-26.
13. Weinerman R, Mainigi M. Why we should transfer frozen instead of fresh embryos: the translational rationale. *Fertil Steril*. 2014;102(1):10-8.
14. von Versen-Höynck F, Schaub AM, Chi YY, Chiu KH, Liu J, Lingis M, et al. Increased Preeclampsia Risk and Reduced Aortic Compliance With In Vitro Fertilization Cycles in the Absence of a Corpus Luteum. *Hypertension*. 2019;73(3):640-9.
15. Mantikou E, Youssef MA, van Wely M, van der Veen F, Al-Inany HG, Repping S, et al. Embryo culture media and IVF/ICSI success rates: a systematic review. *Hum Reprod Update*. 2013;19(3):210-20.
16. Steegers-Theunissen RP, Twigt J, Pestinger V, Sinclair KD. The periconceptional period, reproduction and long-term health of offspring: the importance of one-carbon metabolism. *Hum Reprod Update*. 2013;19(6):640-55.
17. Rizzo G, Aiello E, Pietrolucci ME, Arduini D. Are There Differences in Placental Volume and Uterine Artery Doppler in Pregnancies Resulting From the Transfer of Fresh Versus Frozen-Thawed Embryos Through In Vitro Fertilization. *Reprod Sci*. 2016;23(10):1381-6.
18. Cavoretto PI, Farina A, Girardelli S, Gaeta G, Spinillo S, Morano D, et al. Greater fetal crown-rump length growth with the use of in vitro fertilization or intracytoplasmic sperm injection conceptions after thawed versus fresh blastocyst transfers: secondary analysis of a prospective cohort study. *Fertil Steril*. 2021;116(1):147-56.
19. Shih W, Rushford DD, Bourne H, Garrett C, McBain JC, Healy DL, et al. Factors affecting low birthweight after assisted reproduction technology: difference between transfer of fresh and cryopreserved embryos suggests an adverse effect of oocyte collection. *Hum Reprod*. 2008;23(7):1644-53.

20. Spijkers S, Lens JW, Schats R, Lambalk CB. Fresh and Frozen-Thawed Embryo Transfer Compared to Natural Conception: Differences in Perinatal Outcome. *Gynecol Obstet Invest.* 2017;82(6):538-46.
21. Pinborg A, Loft A, Aaris Henningsen AK, Rasmussen S, Andersen AN. Infant outcome of 957 singletons born after frozen embryo replacement: the Danish National Cohort Study 1995-2006. *Fertil Steril.* 2010;94(4):1320-7.
22. Sazonova A, Källen K, Thurin-Kjellberg A, Wennerholm UB, Bergh C. Obstetric outcome in singletons after in vitro fertilization with cryopreserved/thawed embryos. *Hum Reprod.* 2012;27(5):1343-50.
23. Singh B, Reschke L, Segars J, Baker VL. Frozen-thawed embryo transfer: the potential importance of the corpus luteum in preventing obstetrical complications. *Fertil Steril.* 2020;113(2):252-7.
24. Nakashima A, Araki R, Tani H, Ishihara O, Kuwahara A, Irahara M, et al. Implications of assisted reproductive technologies on term singleton birth weight: an analysis of 25,777 children in the national assisted reproduction registry of Japan. *Fertil Steril.* 2013;99(2):450-5.
25. Choux C, Carmignac V, Bruno C, Sagot P, Vaiman D, Fauque P. The placenta: phenotypic and epigenetic modifications induced by Assisted Reproductive Technologies throughout pregnancy. *Clin Epigenetics.* 2015;7(1):87.
26. McDonald SD, Han Z, Mulla S, Murphy KE, Beyene J, Ohlsson A. Preterm birth and low birth weight among in vitro fertilization singletons: a systematic review and meta-analyses. *Eur J Obstet Gynecol Reprod Biol.* 2009;146(2):138-48.
27. Hansen M, Kurinczuk JJ, Milne E, de Klerk N, Bower C. Assisted reproductive technology and birth defects: a systematic review and meta-analysis. *Hum Reprod Update.* 2013;19(4):330-53.
28. Boulet SL, Kirby RS, Reefhuis J, Zhang Y, Sunderam S, Cohen B, et al. Assisted Reproductive Technology and Birth Defects Among Liveborn Infants in Florida, Massachusetts, and Michigan, 2000-2010. *JAMA Pediatr.* 2016;170(6):e154934.
29. Ishihara O, Araki R, Kuwahara A, Itakura A, Saito H, Adamson GD. Impact of frozen-thawed single-blastocyst transfer on maternal and neonatal outcome: an analysis of 277,042 single-embryo transfer cycles from 2008 to 2010 in Japan. *Fertil Steril.* 2014;101(1):128-33.
30. Maheshwari A, Pandey S, Shetty A, Hamilton M, Bhattacharya S. Obstetric and perinatal outcomes in singleton pregnancies resulting from the transfer of frozen thawed versus fresh embryos generated through in vitro fertilization treatment: a systematic review and meta-analysis. *Fertil Steril.* 2012;98(2):368-77.e1-9.
31. Syngelaki A, Chelemen T, Dagklis T, Allan L, Nicolaides KH. Challenges in the diagnosis of fetal non-chromosomal abnormalities at 11-13 weeks. *Prenat Diagn.* 2011;31(1):90-102.
32. Santorum M, Wright D, Syngelaki A, Karagiotti N, Nicolaides KH. Accuracy of first-trimester combined test in screening for trisomies 21, 18 and 13. *Ultrasound Obstet Gynecol.* 2017;49(6):714-20.
33. Wade DT. Ethics, audit, and research: all shades of grey. *Bmj.* 2005;330(7489):468-71.
34. Salomon LJ, Alfirevic Z, Bilardo CM, Chalouhi GE, Ghi T, Kagan KO, et al. ISUOG practice guidelines: performance of first-trimester fetal ultrasound scan. *Ultrasound Obstet Gynecol.* 2013;41(1):102-13.
35. Papageorgiou AT, Kennedy SH, Salomon LJ, Ohuma EO, Cheikh Ismail L, Barros FC, et al. International standards for early fetal size and pregnancy dating based on ultrasound measurement of crown-rump length in the first trimester of pregnancy. *Ultrasound Obstet Gynecol.* 2014;44(6):641-8.
36. Bukowski R, Smith GC, Malone FD, Ball RH, Nyberg DA, Comstock CH, et al. Human sexual size dimorphism in early pregnancy. *Am J Epidemiol.* 2007;165(10):1216-8.
37. van Uitert EM, Exalto N, Burton GJ, Willemsen SP, Koning AH, Eilers PH, et al. Human embryonic growth trajectories and associations with fetal growth and birthweight. *Hum Reprod.* 2013;28(7):1753-61.
38. Papastefanou I, Souka AP, Pilalis A, Eleftheriades M, Michalitsi V, Kassanos D. First trimester prediction of small- and large-for-gestation neonates by an integrated model incorporating ultrasound parameters, biochemical indices and maternal characteristics. *Acta Obstet Gynecol Scand.* 2012;91(1):104-11.
39. Pardo J, Peled Y, Yogeve Y, Melamed N, Ben-Haroush A. Association of crown-rump length at 11 to 14 weeks' gestation and risk of a large-for-gestational-age neonate. *J Ultrasound Med.* 2010;29(9):1315-9.

40. Mook-Kanamori DO, Steegers EA, Eilers PH, Raat H, Hofman A, Jaddoe VW. Risk factors and outcomes associated with first-trimester fetal growth restriction. *Jama*. 2010;303(6):527-34.
41. Bilagi A, Burke DL, Riley RD, Mills I, Kilby MD, Katie Morris R. Association of maternal serum PAPP-A levels, nuchal translucency and crown-rump length in first trimester with adverse pregnancy outcomes: retrospective cohort study. *Prenat Diagn*. 2017;37(7):705-11.
42. von Versen-Höynck F, Petersen JS, Chi YY, Liu J, Baker VL. First trimester pregnancy ultrasound findings as a function of method of conception in an infertile population. *J Assist Reprod Genet*. 2018;35(5):863-70.
43. Guirgis RR, Alshawaf T, Dave R, Craft IL. Transvaginal crown-rump length measurements of 224 successful pregnancies which resulted from gamete intra-Fallopian transfer or in-vitro fertilization. *Hum Reprod*. 1993;8(11):1933-7.
44. Tunón K, Eik-Nes SH, Grøttum P, Von Düring V, Kahn JA. Gestational age in pregnancies conceived after in vitro fertilization: a comparison between age assessed from oocyte retrieval, crown-rump length and biparietal diameter. *Ultrasound Obstet Gynecol*. 2000;15(1):41-6.
45. Bhattacharya S. Maternal and perinatal outcomes after fresh versus frozen embryo transfer-what is the risk-benefit ratio? *Fertil Steril*. 2016;106(2):241-3.
46. Shapiro BS, Daneshmand ST, Restrepo H, Garner FC, Aguirre M, Hudson C. Matched-cohort comparison of single-embryo transfers in fresh and frozen-thawed embryo transfer cycles. *Fertil Steril*. 2013;99(2):389-92.
47. Shapiro BS, Daneshmand ST, Garner FC, Aguirre M, Hudson C, Thomas S. Evidence of impaired endometrial receptivity after ovarian stimulation for in vitro fertilization: a prospective randomized trial comparing fresh and frozen-thawed embryo transfer in normal responders. *Fertil Steril*. 2011;96(2):344-8.
48. Kansal Kalra S, Ratcliffe SJ, Milman L, Gracia CR, Coutifaris C, Barnhart KT. Perinatal morbidity after in vitro fertilization is lower with frozen embryo transfer. *Fertil Steril*. 2011;95(2):548-53.
49. Zheng Y, Li Z, Xiong M, Luo T, Dong X, Huang B, et al. Hormonal replacement treatment improves clinical pregnancy in frozen-thawed embryos transfer cycles: a retrospective cohort study. *Am J Transl Res*. 2013;6(1):85-90.
50. Terho AM, Pelkonen S, Opdahl S, Romundstad LB, Bergh C, Wennerholm UB, et al. High birth weight and large-for-gestational-age in singletons born after frozen compared to fresh embryo transfer, by gestational week: a Nordic register study from the CoNARTaS group. *Hum Reprod*. 2021;36(4):1083-92.
51. Wennerholm UB, Henningsen AK, Romundstad LB, Bergh C, Pinborg A, Skjaerven R, et al. Perinatal outcomes of children born after frozen-thawed embryo transfer: a Nordic cohort study from the CoNARTaS group. *Hum Reprod*. 2013;28(9):2545-53.
52. Pinborg A, Henningsen AA, Loft A, Malchau SS, Forman J, Andersen AN. Large baby syndrome in singletons born after frozen embryo transfer (FET): is it due to maternal factors or the cryotechnique? *Hum Reprod*. 2014;29(3):618-27.
53. Cavoretto PI, Farina A, Gaeta G, Sigismondi C, Spinillo S, Casiero D, et al. Uterine artery Doppler in singleton pregnancies conceived after in-vitro fertilization or intracytoplasmic sperm injection with fresh vs frozen blastocyst transfer: longitudinal cohort study. *Ultrasound Obstet Gynecol*. 2020;56(4):603-10.
54. Laval M, Garlantézec R, Guivarc'h-Levêque A. Birthweight difference of singletons conceived through in vitro fertilization with frozen versus fresh embryo transfer: An analysis of 5406 embryo transfers in a retrospective study 2013–2018. *Journal of Gynecology Obstetrics and Human Reproduction*. 2020;49(1):101644.
55. Berntsen S, Söderström-Anttila V, Wennerholm UB, Laivuori H, Loft A, Oldereid NB, et al. The health of children conceived by ART: 'the chicken or the egg?'. *Hum Reprod Update*. 2019;25(2):137-58.
56. Bay B, Lyngsø J, Hohwü L, Kesmodel US. Childhood growth of singletons conceived following in vitro fertilisation or intracytoplasmic sperm injection: a systematic review and meta-analysis. *Bjog*. 2019;126(2):158-66.
57. ACOG. Practice Bulletin no. 146: Management of Late-Term and Postterm Pregnancies

Obstet Gynecol 2014 [

58. Martins WP, Nastri CO, Rienzi L, van der Poel SZ, Gracia CR, Racowsky C. Obstetrical and perinatal outcomes following blastocyst transfer compared to cleavage transfer: a systematic review and meta-analysis. *Hum Reprod.* 2016;31(11):2561-9.
59. Zhang J, Du M, Li Z, Wang L, Hu J, Zhao B, et al. Fresh versus frozen embryo transfer for full-term singleton birth: a retrospective cohort study. *J Ovarian Res.* 2018;11(1):59.
60. Ubaldi FM, Cimadomo D, Vaiarelli A, Fabozzi G, Venturella R, Maggiulli R, et al. Advanced Maternal Age in IVF: Still a Challenge? The Present and the Future of Its Treatment. *Front Endocrinol (Lausanne).* 2019;10:94.
61. Leon G, Papetta A, Spiliopoulou C. Overview of the Greek legislation regarding assisted reproduction and comparison with the EU legal framework. *Reprod Biomed Online.* 2011;23(7):820-3.
62. Iuculano A, Stagnati V, Serrenti M, Peddes C, Monni G, Sole G, et al. Crown-rump length: are they different or similar after homologous vs heterologous oocyte/embryo donation? *Am J Obstet Gynecol.* 2017;217(2):224-5.
63. van Loendersloot LL, Moolenaar LM, van Wely M, Repping S, Bossuyt PM, Hompes PGA, et al. Cost-effectiveness of single versus double embryo transfer in IVF in relation to female age. *Eur J Obstet Gynecol Reprod Biol.* 2017;214:25-30.
64. In vitro fertilization and multiple pregnancies: an evidence-based analysis. *Ont Health Technol Assess Ser.* 2006;6(18):1-63.
65. Veleza Z, Vilska S, Hydén-Granskog C, Tiitinen A, Tapanainen JS, Martikainen H. Elective single embryo transfer in women aged 36-39 years. *Hum Reprod.* 2006;21(8):2098-102.
66. Simopoulou M, Sfakianoudis K, Maziotis E, Tsioulou P, Grigoriadis S, Rapani A, et al. PGT-A: who and when? A systematic review and network meta-analysis of RCTs. *J Assist Reprod Genet.* 2021;38(8):1939-57.
67. Turco MY, Moffett A. Development of the human placenta. *Development.* 2019;146(22).
68. Seet EL, Kay HH, Wu S, Terplan M. Placenta accreta: depth of invasion and neonatal outcomes. *J Matern Fetal Neonatal Med.* 2012;25(10):2042-5.
69. Moffat R, Beutler S, Schötzau A, De Geyter M, De Geyter C. Endometrial thickness influences neonatal birth weight in pregnancies with obstetric complications achieved after fresh IVF-ICSI cycles. *Arch Gynecol Obstet.* 2017;296(1):115-22.
70. Choux C, Binquet C, Carmignac V, Bruno C, Chapusot C, Barberet J, et al. The epigenetic control of transposable elements and imprinted genes in newborns is affected by the mode of conception: ART versus spontaneous conception without underlying infertility. *Hum Reprod.* 2018;33(2):331-40.
71. Lee B, Koeppel AF, Wang ET, Gonzalez TL, Sun T, Kroener L, et al. Differential gene expression during placentation in pregnancies conceived with different fertility treatments compared with spontaneous pregnancies. *Fertil Steril.* 2019;111(3):535-46.
72. Marjonen H, Auvinen P, Kahila H, Tšuiiko O, Kōks S, Tiirats A, et al. rs10732516 polymorphism at the IGF2/H19 locus associates with genotype-specific effects on placental DNA methylation and birth weight of newborns conceived by assisted reproductive technology. *Clin Epigenetics.* 2018;10:80.
73. Choux C, Ginod P, Barberet J, Rousseau T, Bruno C, Sagot P, et al. Placental volume and other first-trimester outcomes: are there differences between fresh embryo transfer, frozen-thawed embryo transfer and natural conception? *Reprod Biomed Online.* 2019;38(4):538-48.
74. Hafner E, Metzenbauer M, Höfinger D, Munkel M, Gassner R, Schuchter K, et al. Placental growth from the first to the second trimester of pregnancy in SGA-foetuses and pre-eclamptic pregnancies compared to normal foetuses. *Placenta.* 2003;24(4):336-42.
75. Cavoletto PI, Farina A, Miglio R, Zamagni G, Girardelli S, Vanni VS, et al. Prospective longitudinal cohort study of uterine arteries Doppler in singleton pregnancies obtained by IVF/ICSI with oocyte donation or natural conception. *Hum Reprod.* 2020;35(11):2428-38.
76. Valiño N, Giunta G, Gallo DM, Akolekar R, Nicolaides KH. Uterine artery pulsatility index at 30-34 weeks' gestation in the prediction of adverse perinatal outcome. *Ultrasound Obstet Gynecol.* 2016;47(3):308-15.
77. Paules C, Youssef L, Rovira C, Crovetto F, Nadal A, Peguero A, et al. Distinctive patterns of placental lesions in pre-eclampsia vs small-for-gestational age and their association with fetoplacental Doppler. *Ultrasound Obstet Gynecol.* 2019;54(5):609-16.

78. Ciobanu A, Rouvali A, Syngelaki A, Akolekar R, Nicolaides KH. Prediction of small for gestational age neonates: screening by maternal factors, fetal biometry, and biomarkers at 35-37 weeks' gestation. *Am J Obstet Gynecol.* 2019;220(5):486.e1-.e11.
79. O'Gorman N, Wright D, Poon LC, Rolnik DL, Syngelaki A, Wright A, et al. Accuracy of competing-risks model in screening for pre-eclampsia by maternal factors and biomarkers at 11-13 weeks' gestation. *Ultrasound Obstet Gynecol.* 2017;49(6):751-5.
80. Khalil A, Cowans NJ, Spencer K, Goichman S, Meiri H, Harrington K. First-trimester markers for the prediction of pre-eclampsia in women with a-priori high risk. *Ultrasound Obstet Gynecol.* 2010;35(6):671-9.
81. Velauthar L, Plana MN, Kalidindi M, Zamora J, Thilaganathan B, Illanes SE, et al. First-trimester uterine artery Doppler and adverse pregnancy outcome: a meta-analysis involving 55,974 women. *Ultrasound Obstet Gynecol.* 2014;43(5):500-7.
82. Ferrazzi E, Rigano S, Padoan A, Boito S, Pennati G, Galan HL. Uterine artery blood flow volume in pregnant women with an abnormal pulsatility index of the uterine arteries delivering normal or intrauterine growth restricted newborns. *Placenta.* 2011;32(7):487-92.
83. Nakamura Y, Yaguchi C, Itoh H, Sakamoto R, Kimura T, Furuta N, et al. Morphologic characteristics of the placental basal plate in in vitro fertilization pregnancies: a possible association with the amount of bleeding in delivery. *Hum Pathol.* 2015;46(8):1171-9.
84. Brosens I, Pijnenborg R, Vercruysse L, Romero R. The "Great Obstetrical Syndromes" are associated with disorders of deep placentation. *Am J Obstet Gynecol.* 2011;204(3):193-201.
85. Morgan TK. Role of the Placenta in Preterm Birth: A Review. *Am J Perinatol.* 2016;33(3):258-66.
86. Thilaganathan B. Placental syndromes: getting to the heart of the matter. *Ultrasound Obstet Gynecol.* 2017;49(1):7-9.
87. Shanis DL, Jessmon P, Sinaii N, Armant DR, Stratton P. IVF and increased risk for preeclampsia revisited: a meta-analysis. *Fertility and Sterility.* 2011;96(3):S181.
88. Conrad KP, Baker VL. Corpus luteal contribution to maternal pregnancy physiology and outcomes in assisted reproductive technologies. *Am J Physiol Regul Integr Comp Physiol.* 2013;304(2):R69-72.
89. Shi Y, Sun Y, Hao C, Zhang H, Wei D, Zhang Y, et al. Transfer of Fresh versus Frozen Embryos in Ovulatory Women. *N Engl J Med.* 2018;378(2):126-36.
90. Odibo AO, Goetzinger KR, Huster KM, Christiansen JK, Odibo L, Tuuli MG. Placental volume and vascular flow assessed by 3D power Doppler and adverse pregnancy outcomes. *Placenta.* 2011;32(3):230-4.
91. Green MP, Mouat F, Miles HL, Pleasants AB, Henderson HV, Hofman PL, et al. The phenotype of an IVF child is associated with peri-conception measures of follicular characteristics and embryo quality. *Human Reproduction.* 2014;29(11):2583-91.
92. Viganò P, Alteri A, Busnelli A, Vanni VS, Somigliana E. Frozen IVF Cycles to Circumvent the Hormonal Storm on Endometrium. *Trends Endocrinol Metab.* 2020;31(4):296-307.
93. Rienzi L, Gracia C, Maggiulli R, LaBarbera AR, Kaser DJ, Ubaldi FM, et al. Oocyte, embryo and blastocyst cryopreservation in ART: systematic review and meta-analysis comparing slow-freezing versus vitrification to produce evidence for the development of global guidance. *Hum Reprod Update.* 2017;23(2):139-55.
94. Gu F, Li S, Zheng L, Gu J, Li T, Du H, et al. Perinatal outcomes of singletons following vitrification versus slow-freezing of embryos: a multicenter cohort study using propensity score analysis. *Hum Reprod.* 2019;34(9):1788-98.
95. Delle Piane L, Lin W, Liu X, Donjacour A, Minasi P, Revelli A, et al. Effect of the method of conception and embryo transfer procedure on mid-gestation placenta and fetal development in an IVF mouse model. *Hum Reprod.* 2010;25(8):2039-46.
96. Collier AC, Miyagi SJ, Yamauchi Y, Ward MA. Assisted reproduction technologies impair placental steroid metabolism. *J Steroid Biochem Mol Biol.* 2009;116(1-2):21-8.
97. Farin PW, Piedrahita JA, Farin CE. Errors in development of fetuses and placentas from in vitro-produced bovine embryos. *Theriogenology.* 2006;65(1):178-91.
98. Romundstad LB, Romundstad PR, Sunde A, von Düring V, Skjaerven R, Vatten LJ. Increased risk of placenta previa in pregnancies following IVF/ICSI; a comparison of ART and non-ART pregnancies in the same mother. *Hum Reprod.* 2006;21(9):2353-8.

99. Savasi VM, Mandia L, Laoreti A, Cetin I. Maternal and fetal outcomes in oocyte donation pregnancies. *Hum Reprod Update*. 2016;22(5):620-33.
100. Klatsky PC, Delaney SS, Caughey AB, Tran ND, Schattman GL, Rosenwaks Z. The role of embryonic origin in preeclampsia: a comparison of autologous in vitro fertilization and ovum donor pregnancies. *Obstet Gynecol*. 2010;116(6):1387-92.
101. Blázquez A, García D, Rodríguez A, Vassena R, Figueras F, Vernaev V. Is oocyte donation a risk factor for preeclampsia? A systematic review and meta-analysis. *J Assist Reprod Genet*. 2016;33(7):855-63.
102. Moreno-Sepulveda J, Checa MA. Risk of adverse perinatal outcomes after oocyte donation: a systematic review and meta-analysis. *J Assist Reprod Genet*. 2019;36(10):2017-37.
103. Conrad KP, Petersen JW, Chi YY, Zhai X, Li M, Chiu KH, et al. Maternal Cardiovascular Dysregulation During Early Pregnancy After In Vitro Fertilization Cycles in the Absence of a Corpus Luteum. *Hypertension*. 2019;74(3):705-15.
104. Osol G, Mandala M. Maternal uterine vascular remodeling during pregnancy. *Physiology (Bethesda)*. 2009;24:58-71.
105. Albrecht ED, Pepe GJ. Estrogen regulation of placental angiogenesis and fetal ovarian development during primate pregnancy. *Int J Dev Biol*. 2010;54(2-3):397-408.
106. Aberdeen GW, Bonagura TW, Harman CR, Pepe GJ, Albrecht ED. Suppression of trophoblast uterine spiral artery remodeling by estrogen during baboon pregnancy: impact on uterine and fetal blood flow dynamics. *Am J Physiol Heart Circ Physiol*. 2012;302(10):H1936-44.
107. Horcujadas JA, Riesewijk A, Polman J, van Os R, Pellicer A, Mosselman S, et al. Effect of controlled ovarian hyperstimulation in IVF on endometrial gene expression profiles. *Mol Hum Reprod*. 2005;11(3):195-205.
108. Kolb BA, Paulson RJ. The luteal phase of cycles utilizing controlled ovarian hyperstimulation and the possible impact of this hyperstimulation on embryo implantation. *Am J Obstet Gynecol*. 1997;176(6):1262-7; discussion 7-9.
109. Papanikolaou EG, Bourgain C, Kolibianakis E, Tournaye H, Devroey P. Steroid receptor expression in late follicular phase endometrium in GnRH antagonist IVF cycles is already altered, indicating initiation of early luteal phase transformation in the absence of secretory changes. *Hum Reprod*. 2005;20(6):1541-7.
110. Horcujadas JA, Díaz-Gimeno P, Pellicer A, Simón C. Uterine receptivity and the ramifications of ovarian stimulation on endometrial function. *Semin Reprod Med*. 2007;25(6):454-60.
111. Conrad KP. Maternal vasodilation in pregnancy: the emerging role of relaxin. *Am J Physiol Regul Integr Comp Physiol*. 2011;301(2):R267-75.
112. Ginström Ernstad E, Spangmose AL, Opdahl S, Henningsen AA, Romundstad LB, Tiitinen A, et al. Perinatal and maternal outcome after vitrification of blastocysts: a Nordic study in singletons from the CoNARTaS group. *Hum Reprod*. 2019;34(11):2282-9.
113. Pereira N, Elias RT, Christos PJ, Petrini AC, Hancock K, Lekovich JP, et al. Supraphysiologic estradiol is an independent predictor of low birth weight in full-term singletons born after fresh embryo transfer. *Hum Reprod*. 2017;32(7):1410-7.
114. Ginod P, Choux C, Barberet J, Rousseau T, Bruno C, Khallouk B, et al. Singleton fetal growth kinetics depend on the mode of conception. *Fertility and Sterility*. 2018;110(6):1109-17.e2.
115. Nastri CO, Lensen SF, Gibreel A, Raine-Fenning N, Ferriani RA, Bhattacharya S, et al. Endometrial injury in women undergoing assisted reproductive techniques. *Cochrane Database Syst Rev*. 2015(3):Cd009517.
116. Jauniaux E, Jurkovic D. Placenta accreta: pathogenesis of a 20th century iatrogenic uterine disease. *Placenta*. 2012;33(4):244-51.
117. Imudia AN, Awonuga AO, Doyle JO, Kaimal AJ, Wright DL, Toth TL, et al. Peak serum estradiol level during controlled ovarian hyperstimulation is associated with increased risk of small for gestational age and preeclampsia in singleton pregnancies after in vitro fertilization. *Fertil Steril*. 2012;97(6):1374-9.
118. Roos N, Kieler H, Sahlin L, Ekman-Ordeberg G, Falconer H, Stephansson O. Risk of adverse pregnancy outcomes in women with polycystic ovary syndrome: population based cohort study. *Bmj*. 2011;343:d6309.
119. Järvelä IY, Pelkonen S, Uimari O, Mäkilä K, Puukka K, Ruokonen A, et al. Controlled ovarian hyperstimulation leads to high progesterone and estradiol levels during early pregnancy. *Hum Reprod*. 2014;29(11):2393-401.

120. van Duijn L, Rousian M, Reijnders IF, Willemsen SP, Baart EB, Laven JSE, et al. The influence of frozen-thawed and fresh embryo transfer on utero-placental (vascular) development: the Rotterdam Periconception cohort. *Hum Reprod.* 2021;36(8):2091-100.
121. Wang X, Du M, Guan Y, Wang B, Zhang J, Liu Z. Comparative neonatal outcomes in singleton births from blastocyst transfers or cleavage-stage embryo transfers: a systematic review and meta-analysis. *Reprod Biol Endocrinol.* 2017;15(1):36.
122. Alviggi C, Conforti A, Carbone IF, Borrelli R, de Placido G, Guerriero S. Influence of cryopreservation on perinatal outcome after blastocyst- vs cleavage-stage embryo transfer: systematic review and meta-analysis. *Ultrasound in Obstetrics & Gynecology.* 2018;51(1):54-63.
123. Nelissen EC, van Montfoort AP, Dumoulin JC, Evers JL. Epigenetics and the placenta. *Hum Reprod Update.* 2011;17(3):397-417.
124. Market-Velker BA, Zhang L, Magri LS, Bonvissuto AC, Mann MR. Dual effects of superovulation: loss of maternal and paternal imprinted methylation in a dose-dependent manner. *Hum Mol Genet.* 2010;19(1):36-51.
125. Bernard A, Fuller BJ. Cryopreservation of human oocytes: a review of current problems and perspectives. *Hum Reprod Update.* 1996;2(3):193-207.
126. Van der Elst J, Van den Abbeel E, Jacobs R, Wisse E, Van Steirteghem A. Effect of 1,2-propanediol and dimethylsulphoxide on the meiotic spindle of the mouse oocyte. *Hum Reprod.* 1988;3(8):960-7.
127. Bonne S, Sauleau E, Sananes N, Akaladios C, Rongières C, Pirrello O. Influence of medically assisted reproduction techniques on crown-rump length and biochemical markers of trisomy 21 in the first trimester of pregnancy. *Fertil Steril.* 2016;105(2):410-6.
128. Knight B, Brereton A, Powell RJ, Liversedge H. Assessing the accuracy of ultrasound estimation of gestational age during routine antenatal care in in vitro fertilization (IVF) pregnancies. *Ultrasound.* 2018;26(1):49-53.
129. Senapati S, Wang F, Ord T, Coutifaris C, Feng R, Mainigi M. Superovulation alters the expression of endometrial genes critical to tissue remodeling and placentation. *J Assist Reprod Genet.* 2018;35(10):1799-808.
130. Cooper AR, O'Neill KE, Allsworth JE, Jungheim ES, Odibo AO, Gray DL, et al. Smaller fetal size in singletons after infertility therapies: the influence of technology and the underlying infertility. *Fertil Steril.* 2011;96(5):1100-6.
131. Deligdisch L. Hormonal pathology of the endometrium. *Mod Pathol.* 2000;13(3):285-94.
132. Cavorotto P, Dallagiovanna C, Viganò P, Somigliana E, Persico N, Papaleo E, et al. First trimester combined screening test in pregnancies derived from blastocyst transfer. *European Journal of Obstetrics and Gynecology and Reproductive Biology.* 2016;198:50-5.
133. Snijders RJ, Noble P, Sebire N, Souka A, Nicolaides KH. UK multicentre project on assessment of risk of trisomy 21 by maternal age and fetal nuchal-translucency thickness at 10-14 weeks of gestation. Fetal Medicine Foundation First Trimester Screening Group. *Lancet.* 1998;352(9125):343-6.
134. von Kaisenberg CS, Nicolaides KH, Brand-Saberi B. Lymphatic vessel hypoplasia in fetuses with Turner syndrome. *Hum Reprod.* 1999;14(3):823-6.
135. Sathananthan AH, Trounson A, Freemann L, Brady T. The effects of cooling human oocytes. *Hum Reprod.* 1988;3(8):968-77.
136. Barkai G, Goldman B, Ries L, Chaki R, Dor J, Cuckle H. Down's syndrome screening marker levels following assisted reproduction. *Prenat Diagn.* 1996;16(12):1111-4.
137. Hui PW, Tang MH, Lam YH, Ng EH, Yeung WS, Ho PC. Maternal serum hCG and alpha-fetoprotein levels in pregnancies conceived after IVF or ICSI with fresh and frozen-thawed embryos. *Hum Reprod.* 2003;18(3):572-5.
138. Plourde D, Vigneault C, Lemay A, Breton L, Gagné D, Laflamme I, et al. Contribution of oocyte source and culture conditions to phenotypic and transcriptomic variation in commercially produced bovine blastocysts. *Theriogenology.* 2012;78(1):116-31.e1-3.
139. Chen M, Wu L, Zhao J, Wu F, Davies MJ, Wittert GA, et al. Altered glucose metabolism in mouse and humans conceived by IVF. *Diabetes.* 2014;63(10):3189-98.
140. Barker DJ. Maternal nutrition, fetal nutrition, and disease in later life. *Nutrition.* 1997;13(9):807-13.

141. Green MP, Mouat F, Miles HL, Pleasants AB, Henderson HV, Hofman PL, et al. The phenotype of an IVF child is associated with peri-conception measures of follicular characteristics and embryo quality. *Hum Reprod.* 2014;29(11):2583-91.
142. Raga F, Casañ EM, Kruessel J, Wen Y, Bonilla-Musoles F, Polan ML. The role of gonadotropin-releasing hormone in murine preimplantation embryonic development. *Endocrinology.* 1999;140(8):3705-12.
143. Gründker C, Günthert AR, Millar RP, Emons G. Expression of gonadotropin-releasing hormone II (GnRH-II) receptor in human endometrial and ovarian cancer cells and effects of GnRH-II on tumor cell proliferation. *J Clin Endocrinol Metab.* 2002;87(3):1427-30.
144. Ceelen M, van Weissenbruch MM, Prein J, Smit JJ, Vermeiden JP, Spreeuwenberg M, et al. Growth during infancy and early childhood in relation to blood pressure and body fat measures at age 8-18 years of IVF children and spontaneously conceived controls born to subfertile parents. *Hum Reprod.* 2009;24(11):2788-95.
145. Grace KS, Sinclair KD. Assisted reproductive technology, epigenetics, and long-term health: a developmental time bomb still ticking. *Semin Reprod Med.* 2009;27(5):409-16.
146. Young LE, Sinclair KD, Wilmut I. Large offspring syndrome in cattle and sheep. *Rev Reprod.* 1998;3(3):155-63.
147. Chen Z, Robbins KM, Wells KD, Rivera RM. Large offspring syndrome: a bovine model for the human loss-of-imprinting overgrowth syndrome Beckwith-Wiedemann. *Epigenetics.* 2013;8(6):591-601.
148. Barker DJ. The origins of the developmental origins theory. *J Intern Med.* 2007;261(5):412-7.
149. Barker DJ, Osmond C, Kajantie E, Eriksson JG. Growth and chronic disease: findings in the Helsinki Birth Cohort. *Ann Hum Biol.* 2009;36(5):445-58.
150. Kapral N, Miller SE, Scharf RJ, Gurka MJ, DeBoer MD. Associations between birthweight and overweight and obesity in school-age children. *Pediatr Obes.* 2018;13(6):333-41.
151. Norris SA, Osmond C, Gigante D, Kuzawa CW, Ramakrishnan L, Lee NR, et al. Size at birth, weight gain in infancy and childhood, and adult diabetes risk in five low- or middle-income country birth cohorts. *Diabetes Care.* 2012;35(1):72-9.
152. Harder T, Rodekamp E, Schellong K, Dudenhausen JW, Plagemann A. Birth weight and subsequent risk of type 2 diabetes: a meta-analysis. *Am J Epidemiol.* 2007;165(8):849-57.
153. Luyckx VA, Bertram JF, Brenner BM, Fall C, Hoy WE, Ozanne SE, et al. Effect of fetal and child health on kidney development and long-term risk of hypertension and kidney disease. *Lancet.* 2013;382(9888):273-83.
154. Mu M, Wang SF, Sheng J, Zhao Y, Li HZ, Hu CL, et al. Birth weight and subsequent blood pressure: a meta-analysis. *Arch Cardiovasc Dis.* 2012;105(2):99-113.
155. Skilton MR, Siitonen N, Würtz P, Viikari JS, Juonala M, Seppälä I, et al. High birth weight is associated with obesity and increased carotid wall thickness in young adults: the cardiovascular risk in young Finns study. *Arterioscler Thromb Vasc Biol.* 2014;34(5):1064-8.
156. Hutt KJ, Albertini DF. An oocentric view of folliculogenesis and embryogenesis. *Reprod Biomed Online.* 2007;14(6):758-64.
157. Ménéz Y, Dale B, Cohen M. DNA damage and repair in human oocytes and embryos: a review. *Zygote.* 2010;18(4):357-65.
158. Pelkonen S, Hartikainen AL, Ritvanen A, Koivunen R, Martikainen H, Gissler M, et al. Major congenital anomalies in children born after frozen embryo transfer: a cohort study 1995-2006. *Hum Reprod.* 2014;29(7):1552-7.
159. Belva F, Henriët S, Van den Abbeel E, Camus M, Devroey P, Van der Elst J, et al. Neonatal outcome of 937 children born after transfer of cryopreserved embryos obtained by ICSI and IVF and comparison with outcome data of fresh ICSI and IVF cycles. *Hum Reprod.* 2008;23(10):2227-38.
160. Group ECW. Birth defects and congenital health risks in children conceived through assisted reproduction technology (ART): a meeting report. *Journal of Assisted Reproduction and Genetics.* 2014;31(8):947-58.
161. Harvey W. *Exercitationes de Generatione Animalium Londinii* 1651.
162. Hutt KJ, Shi Z, Petroff BK, Albertini DF. The environmental toxicant 2,3,7,8-tetrachlorodibenzo-p-dioxin disturbs the establishment and maintenance of cell polarity in preimplantation rat embryos. *Biol Reprod.* 2010;82(5):914-20.
163. McGinnis LK, Limback SD, Albertini DF. Signaling modalities during oogenesis in mammals. *Curr Top Dev Biol.* 2013;102:227-42.

164. Feng S, Jacobsen SE, Reik W. Epigenetic reprogramming in plant and animal development. *Science*. 2010;330(6004):622-7.
165. Inbar-Feigenberg M, Choufani S, Butcher DT, Roifman M, Weksberg R. Basic concepts of epigenetics. *Fertil Steril*. 2013;99(3):607-15.
166. Horsthemke B. Mechanisms of imprint dysregulation. *Am J Med Genet C Semin Med Genet*. 2010;154c(3):321-8.
167. Jin SC, Homsy J, Zaidi S, Lu Q, Morton S, DePalma SR, et al. Contribution of rare inherited and de novo variants in 2,871 congenital heart disease probands. *Nat Genet*. 2017;49(11):1593-601.
168. Wang C, Lv H, Ling X, Li H, Diao F, Dai J, et al. Association of assisted reproductive technology, germline de novo mutations and congenital heart defects in a prospective birth cohort study. *Cell Res*. 2021;31(8):919-28.
169. Shimada M, Terada T. FSH and LH induce progesterone production and progesterone receptor synthesis in cumulus cells: a requirement for meiotic resumption in porcine oocytes. *Molecular Human Reproduction*. 2002;8(7):612-8.
170. Schachter M, Friedler S, Ron-El R, Zimmerman AL, Strassburger D, Bern O, et al. Can pregnancy rate be improved in gonadotropin-releasing hormone (GnRH) antagonist cycles by administering GnRH agonist before oocyte retrieval? A prospective, randomized study. *Fertil Steril*. 2008;90(4):1087-93.
171. Goldmann JM, Seplyarskiy VB, Wong WSW, Vilboux T, Neerinx PB, Bodian DL, et al. Germline de novo mutation clusters arise during oocyte aging in genomic regions with high double-strand-break incidence. *Nat Genet*. 2018;50(4):487-92.
172. Jiang Z, Wang Y, Lin J, Xu J, Ding G, Huang H. Genetic and epigenetic risks of assisted reproduction. *Best Pract Res Clin Obstet Gynaecol*. 2017;44:90-104.
173. El-Danasouri I, Sterzik K, Rinaldi L, Pacchiarotti A, DeSanto M, Selman H. Effect of transferring a morphologically impaired embryo with a good quality embryo on the pregnancy and implantation rates. *Eur Rev Med Pharmacol Sci*. 2016;20(3):394-8.
174. Brosens JJ, Salker MS, Teklenburg G, Nautiyal J, Salter S, Lucas ES, et al. Uterine selection of human embryos at implantation. *Sci Rep*. 2014;4:3894.
175. Li G, Yu Y, Fan Y, Li C, Xu X, Duan J, et al. Genome wide abnormal DNA methylome of human blastocyst in assisted reproductive technology. *J Genet Genomics*. 2017;44(10):475-81.
176. Licciardi F, McCaffrey C, Oh C, Schmidt-Sarosi C, McCulloh DH. Birth weight is associated with inner cell mass grade of blastocysts. *Fertil Steril*. 2015;103(2):382-7.e2.
177. Rosen MP, Shen S, Dobson AT, Rinaudo PF, McCulloch CE, Cedars MI. A quantitative assessment of follicle size on oocyte developmental competence. *Fertil Steril*. 2008;90(3):684-90.
178. Weimar CH, Macklon NS, Post Uiterweer ED, Brosens JJ, Gellersen B. The motile and invasive capacity of human endometrial stromal cells: implications for normal and impaired reproductive function. *Hum Reprod Update*. 2013;19(5):542-57.
179. Lane M, Gardner DK. Understanding cellular disruptions during early embryo development that perturb viability and fetal development. *Reprod Fertil Dev*. 2005;17(3):371-8.
180. Bloomfield FH, Oliver MH, Harding JE. The late effects of fetal growth patterns. *Arch Dis Child Fetal Neonatal Ed*. 2006;91(4):F299-304.
181. Roseboom TJ, Painter RC, van Abeelen AF, Veenendaal MV, de Rooij SR. Hungry in the womb: what are the consequences? Lessons from the Dutch famine. *Maturitas*. 2011;70(2):141-5.
182. Rumball CW, Bloomfield FH, Oliver MH, Harding JE. Different periods of periconceptional undernutrition have different effects on growth, metabolic and endocrine status in fetal sheep. *Pediatr Res*. 2009;66(6):605-13.
183. Kwong WY, Wild AE, Roberts P, Willis AC, Fleming TP. Maternal undernutrition during the preimplantation period of rat development causes blastocyst abnormalities and programming of postnatal hypertension. *Development*. 2000;127(19):4195-202.
184. Kwong WY, Miller DJ, Ursell E, Wild AE, Wilkins AP, Osmond C, et al. Imprinted gene expression in the rat embryo-fetal axis is altered in response to periconceptional maternal low protein diet. *Reproduction*. 2006;132(2):265-77.
185. Watkins AJ, Ursell E, Panton R, Papenbrock T, Hollis L, Cunningham C, et al. Adaptive responses by mouse early embryos to maternal diet protect fetal growth but predispose to adult onset disease. *Biol Reprod*. 2008;78(2):299-306.

186. Fleming TP, Velazquez MA, Eckert JJ, Lucas ES, Watkins AJ. Nutrition of females during the peri-conceptional period and effects on foetal programming and health of offspring. *Anim Reprod Sci.* 2012;130(3-4):193-7.
187. Lucas E. Epigenetic effects on the embryo as a result of periconceptional environment and assisted reproduction technology. *Reprod Biomed Online.* 2013;27(5):477-85.
188. Eskild A, Monkerud L, Tanbo T. Birthweight and placental weight; do changes in culture media used for IVF matter? Comparisons with spontaneous pregnancies in the corresponding time periods. *Hum Reprod.* 2013;28(12):3207-14.
189. Zhu J, Lin S, Li M, Chen L, Lian Y, Liu P, et al. Effect of in vitro culture period on birthweight of singleton newborns. *Hum Reprod.* 2014;29(3):448-54.
190. Kleijkers SH, Mantikou E, Slappendel E, Consten D, van Echten-Arends J, Wetzels AM, et al. Influence of embryo culture medium (G5 and HTF) on pregnancy and perinatal outcome after IVF: a multicenter RCT. *Hum Reprod.* 2016;31(10):2219-30.
191. Castillo CM, Horne G, Fitzgerald CT, Johnstone ED, Brison DR, Roberts SA. The impact of IVF on birthweight from 1991 to 2015: a cross-sectional study. *Hum Reprod.* 2019;34(5):920-31.
192. Nelissen EC, Van Montfoort AP, Coonen E, Derhaag JG, Geraedts JP, Smits LJ, et al. Further evidence that culture media affect perinatal outcome: findings after transfer of fresh and cryopreserved embryos. *Hum Reprod.* 2012;27(7):1966-76.
193. Kleijkers SH, van Montfoort AP, Smits LJ, Viechtbauer W, Roseboom TJ, Nelissen EC, et al. IVF culture medium affects post-natal weight in humans during the first 2 years of life. *Hum Reprod.* 2014;29(4):661-9.
194. Bloise E, Feuer SK, Rinaudo PF. Comparative intrauterine development and placental function of ART concepti: implications for human reproductive medicine and animal breeding. *Hum Reprod Update.* 2014;20(6):822-39.
195. van Montfoort AP, Hanssen LL, de Sutter P, Viville S, Geraedts JP, de Boer P. Assisted reproduction treatment and epigenetic inheritance. *Hum Reprod Update.* 2012;18(2):171-97.
196. Kleijkers SH, Eijssen LM, Coonen E, Derhaag JG, Mantikou E, Jonker MJ, et al. Differences in gene expression profiles between human preimplantation embryos cultured in two different IVF culture media. *Hum Reprod.* 2015;30(10):2303-11.
197. Fauque P, Mondon F, Letourneur F, Ripoché MA, Journot L, Barbaux S, et al. In vitro fertilization and embryo culture strongly impact the placental transcriptome in the mouse model. *PLoS One.* 2010;5(2):e9218.
198. Mantikou E, Jonker MJ, Wong KM, van Montfoort AP, de Jong M, Breit TM, et al. Factors affecting the gene expression of in vitro cultured human preimplantation embryos. *Hum Reprod.* 2016;31(2):298-311.
199. Nelissen EC, Van Montfoort AP, Smits LJ, Menheere PP, Evers JL, Coonen E, et al. IVF culture medium affects human intrauterine growth as early as the second trimester of pregnancy. *Hum Reprod.* 2013;28(8):2067-74.
200. Cole LA. Hyperglycosylated hCG, a review. *Placenta.* 2010;31(8):653-64.
201. Orasanu B, Jackson KV, Hornstein MD, Racowsky C. Effects of culture medium on HCG concentrations and their value in predicting successful IVF outcome. *Reprod Biomed Online.* 2006;12(5):590-8.
202. Summers MC, Biggers JD. Chemically defined media and the culture of mammalian preimplantation embryos: historical perspective and current issues. *Hum Reprod Update.* 2003;9(6):557-82.
203. Bontekoe S, Heineman MJ, Johnson N, Blake D. Adherence compounds in embryo transfer media for assisted reproductive technologies. *Cochrane Database Syst Rev.* 2014;2014(2):Cd007421.
204. Meintjes M, Chantilis SJ, Ward DC, Douglas JD, Rodriguez AJ, Guerami AR, et al. A randomized controlled study of human serum albumin and serum substitute supplement as protein supplements for IVF culture and the effect on live birth rates. *Hum Reprod.* 2009;24(4):782-9.
205. Sunde A, Brison D, Dumoulin J, Harper J, Lundin K, Magli MC, et al. Time to take human embryo culture seriously. *Hum Reprod.* 2016;31(10):2174-82.
206. Zhang J, Wang Y, Liu H, Mao X, Chen Q, Fan Y, et al. Effect of in vitro culture period on birth weight after vitrified-warmed transfer cycles: analysis of 4,201 singleton newborns. *Fertil Steril.* 2019;111(1):97-104.

207. Coetzee K, Ozgur K, Bulut H, Berkkanoglu M, Humaidan P. Large-for-gestational age is male-gender dependent in artificial frozen embryo transfers cycles: a cohort study of 1295 singleton live births. *Reprod Biomed Online*. 2020;40(1):134-41.
208. Litzky JF, Boulet SL, Esfandiari N, Zhang Y, Kissin DM, Theiler RN, et al. Effect of frozen/thawed embryo transfer on birthweight, macrosomia, and low birthweight rates in US singleton infants. *Am J Obstet Gynecol*. 2018;218(4):433.e1-.e10.
209. Cobo A, Serra V, Garrido N, Olmo I, Pellicer A, Remohí J. Obstetric and perinatal outcome of babies born from vitrified oocytes. *Fertil Steril*. 2014;102(4):1006-15.e4.